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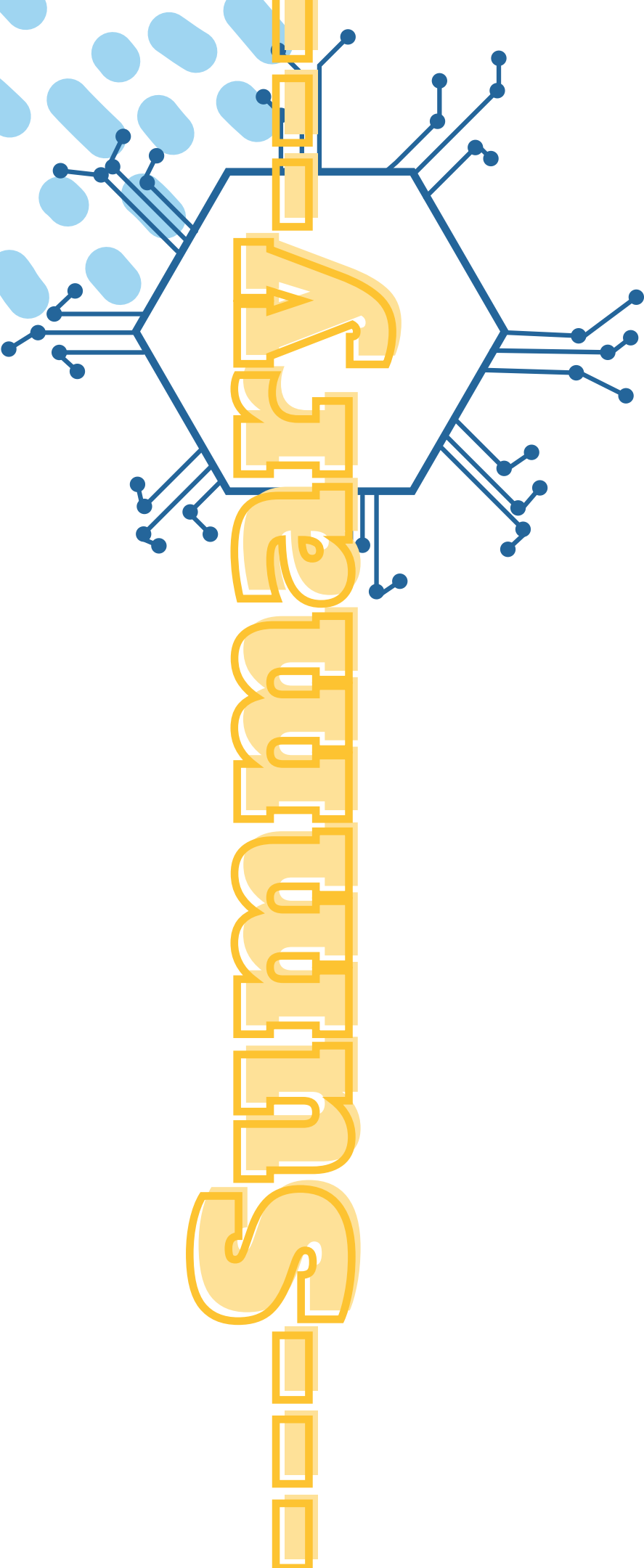


Classification of Alexandrium minutum Occurrences and Blooms in Maritime Context

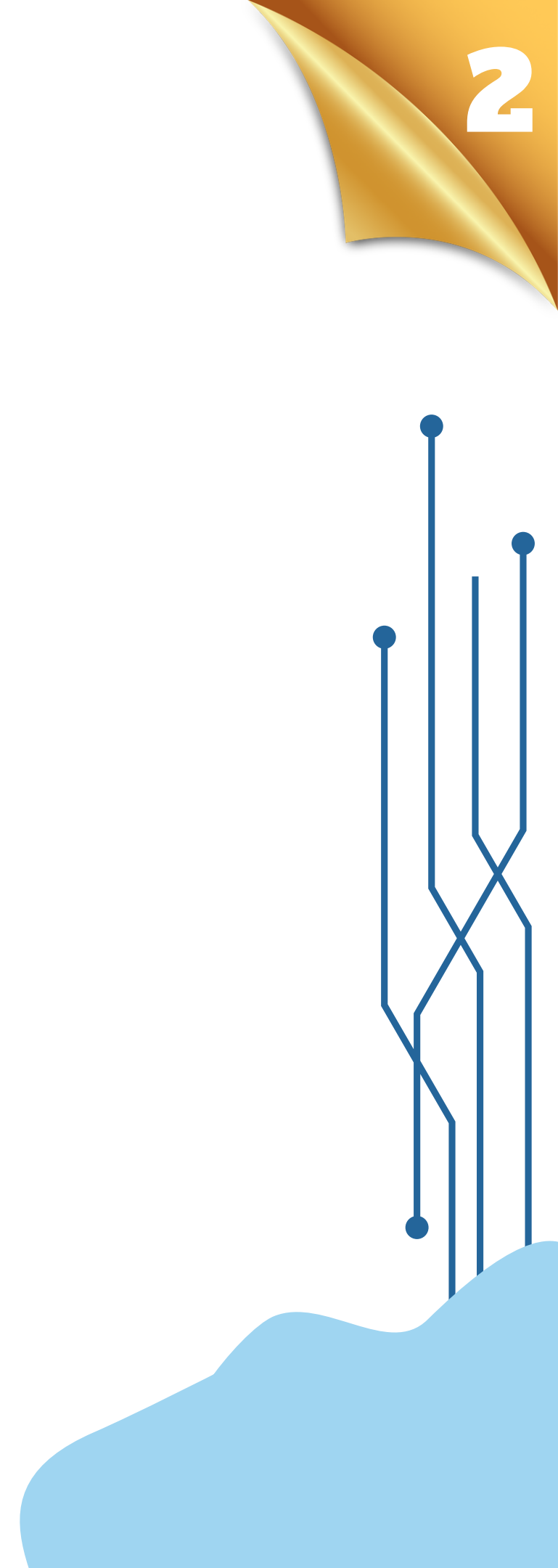
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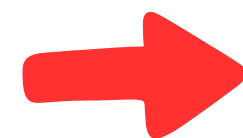
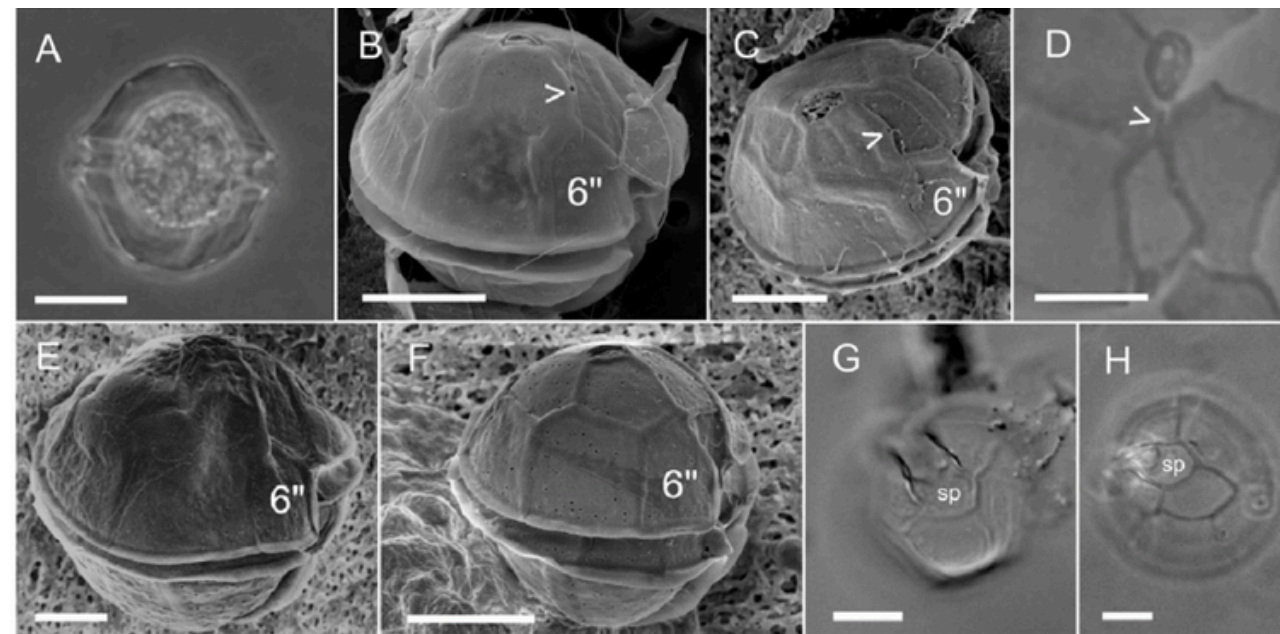
- ✦ Introduction
- ✦ Proposed Method
- ✦ Results and Discussion
- ✦ Conclusion and Perspectives



GENERAL INTRODUCTION

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- ✦ Harmful algal blooms (HABs) are increasing globally and pose a major threat to marine life, fisheries, and human health.
- ✦ *Alexandrium minutum* is a toxic microalga that produces paralytic shellfish toxins.

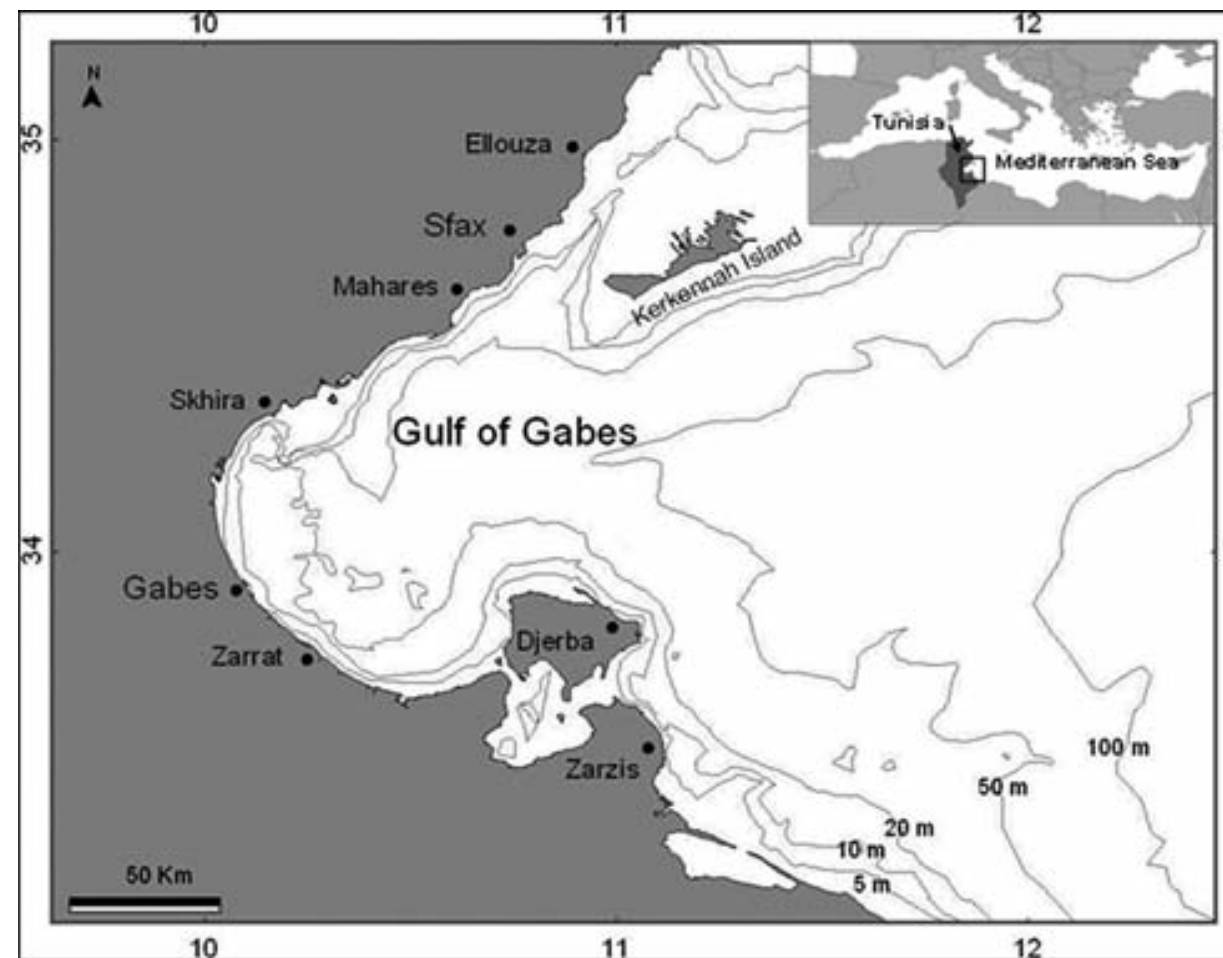


Monitoring and early detection are *essential* to prevent ecological and socio-economic damage.

PROJECT CONTEXT

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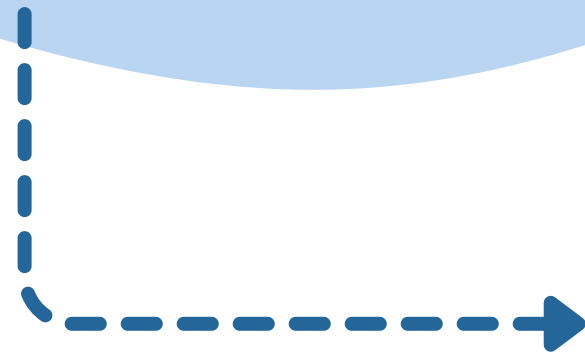
- ✦ Tunisia's coastlines, especially the Gulf of Gabès and northern lagoons, are high-risk bloom zones.
- ✦ Blooms are increasing due to climate change, runoff, and urbanization.



Traditional detection is *slow*, *expert-dependent*, and *not scalable*.

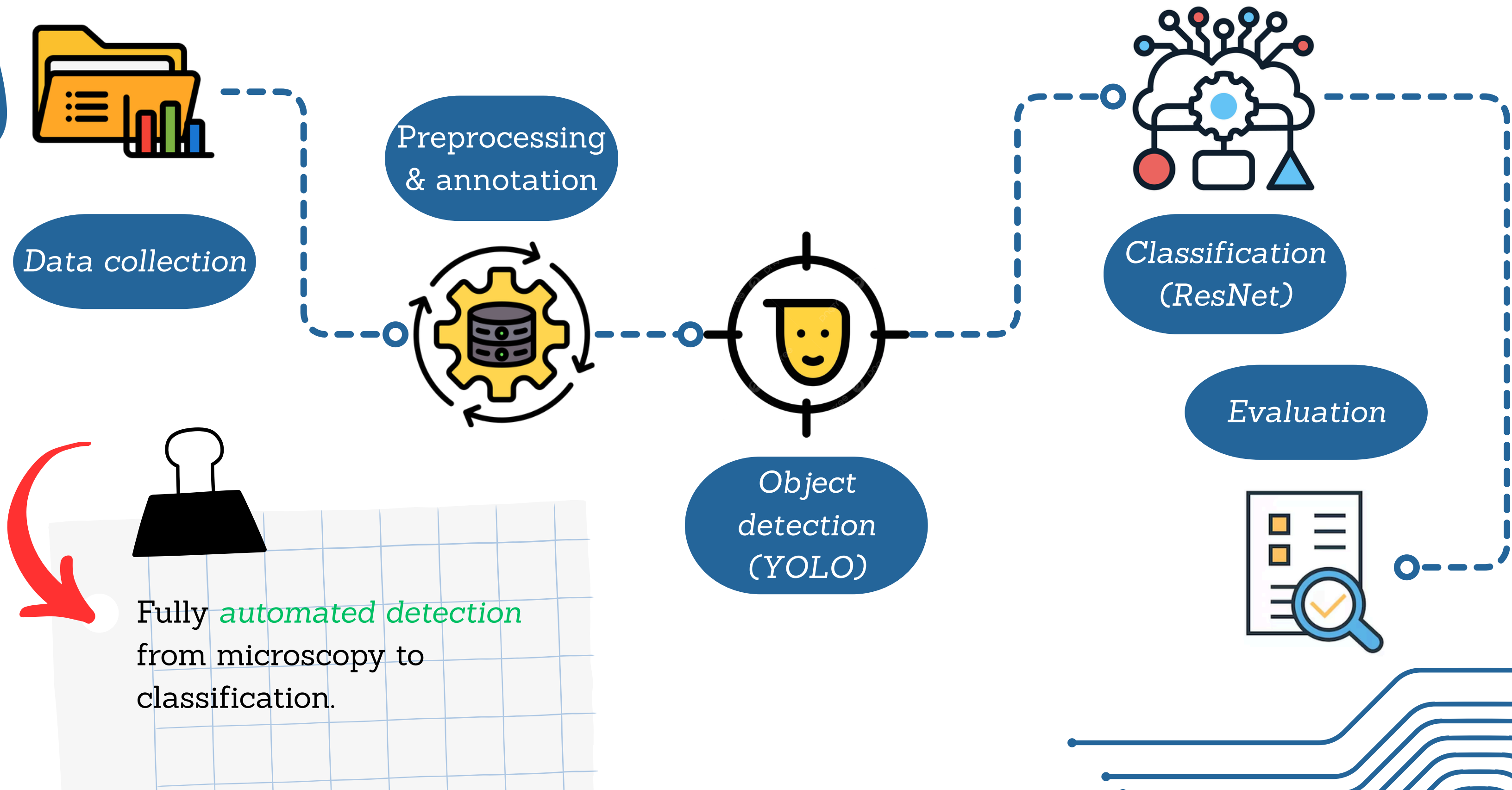
PROBLEM STATEMENT & OBJECTIVES

- ✦ *Lack of annotated data*
- ✦ *Morphological similarity with other species*
- ✦ *Slow/manual identification methods*

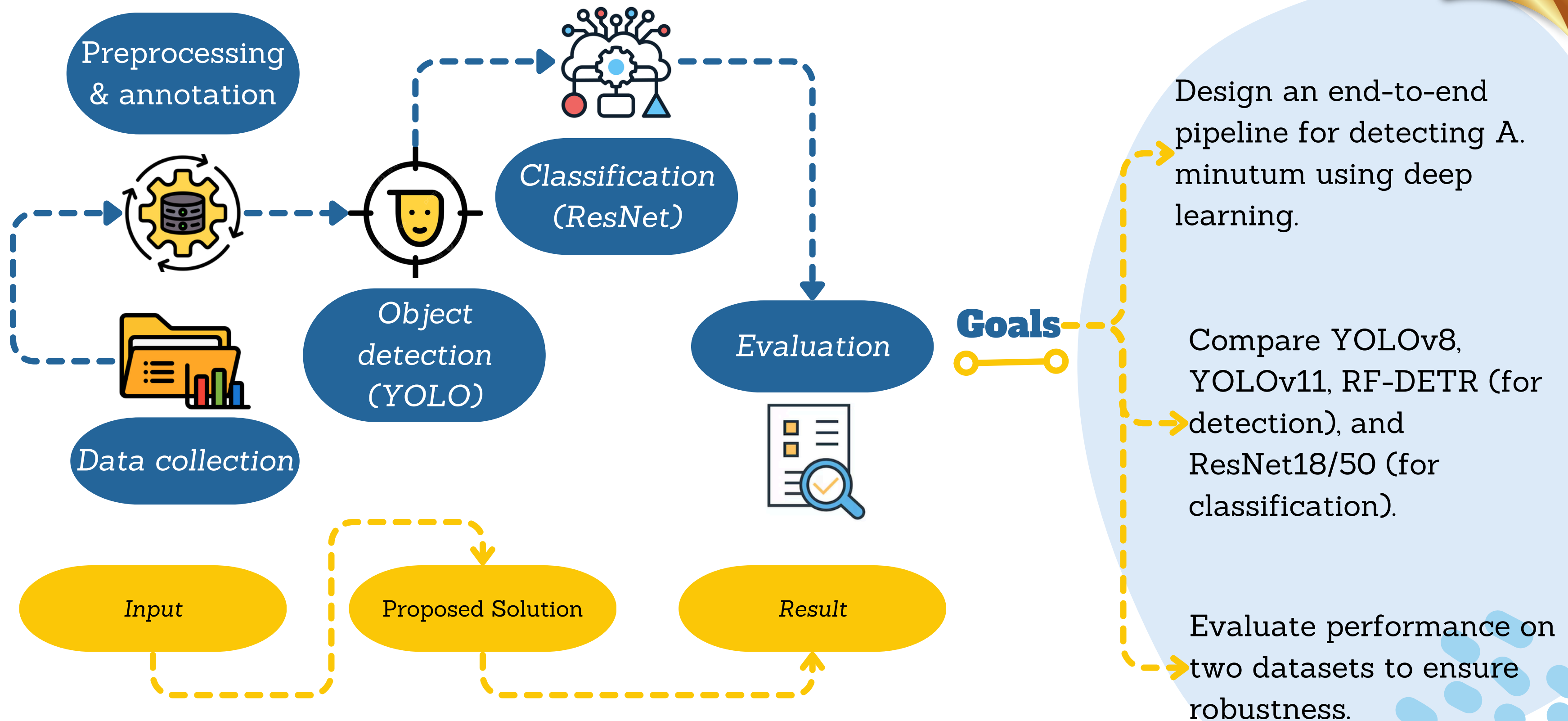


- ✦ Build an AI-based pipeline for real-time detection
- ✦ Test multiple models: YOLOv8, YOLOv11, RF-DETR
- ✦ Enhance monitoring accuracy using deep learning

GENERAL FRAMEWORK



PROPOSED METHOD OBJECTIVES



DATASET & PREPARATION

Dataset 1: Small set collected from web (47 images)

47 images (23 *A. minutum*, 24 others) from Roscoff, CCAP, GBIF.

Preprocessing: grayscale, resize (640x640), auto-orient.

Augmentations: flips, $\pm 15^\circ$ rotations, $\pm 18\%$ zoom.

Final: 113 annotated images (YOLO format) using Roboflow.



CNRS • SORBONNE UNIVERSITÉ
Station Biologique
de Roscoff



GBIF
Global Biodiversity
Information Facility



**Culture
Collection**
of algae & protozoa

Figure 1: Different sources of images for the first Dataset

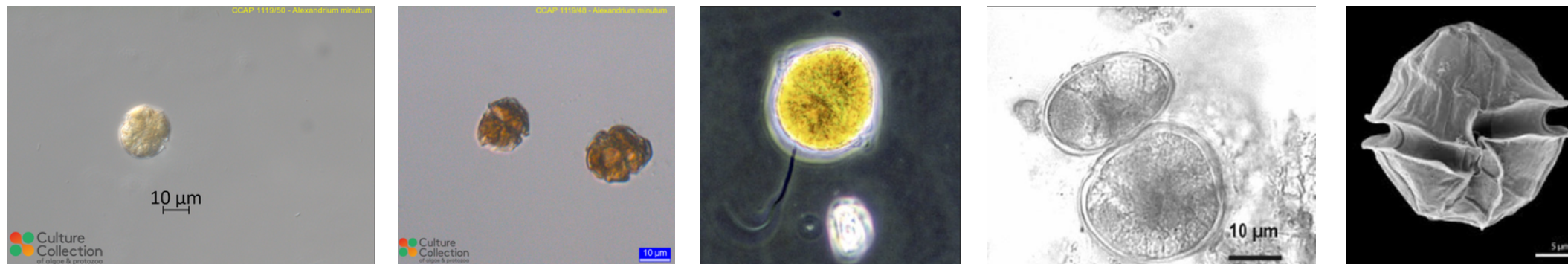


Figure 1: Sample Images from the second Dataset

DATASET & PREPARATION

Dataset 2: Large cytometry dataset (3,766 images) from INSTM

[*] *INSTM*: Institut National des Sciences et Technologies de la Mer.



3,766 images

**2 classes,
7 categories**

→ *Alexandrium Minutum*

→ *Alexandrium Fundyense*

→ *Alexandrium Tamarense*

→ *Gymnodinium Catenatum*

→ *Prorocentrum Lima*

→ *Gonyaulax Polyedra*

→ *Karenia Selliformis*

Other Cells

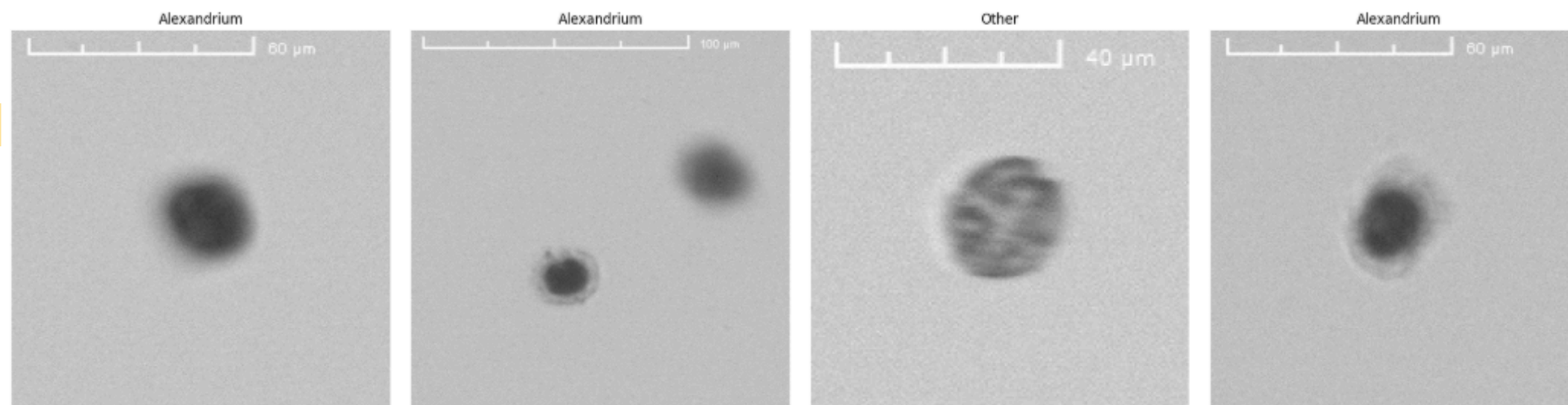
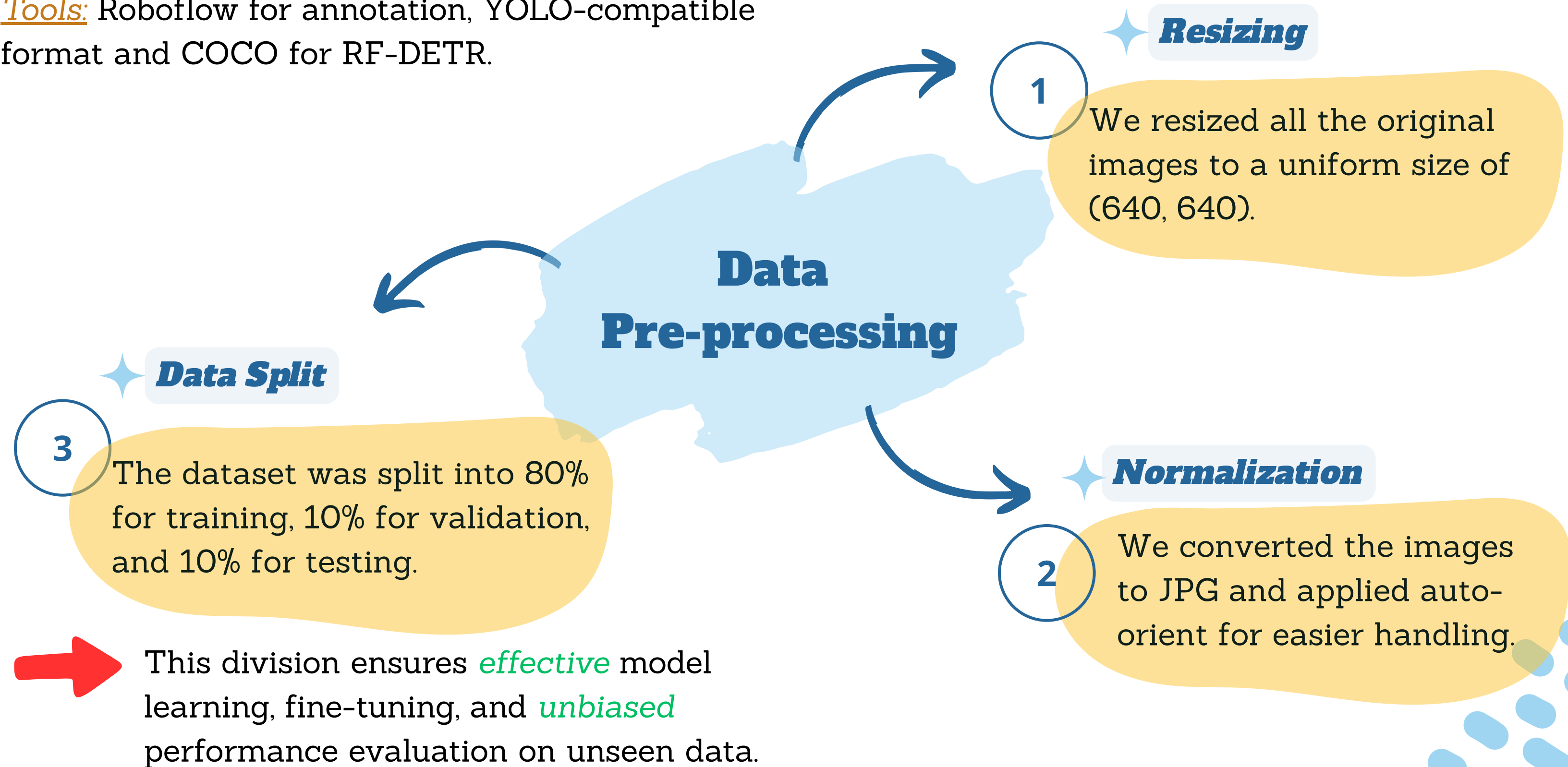


Figure 1: Sample Images from the second Dataset

DATA PREPARATION

Tools: Roboflow for annotation, YOLO-compatible format and COCO for RF-DETR.



MODELS USED

Detection Models:

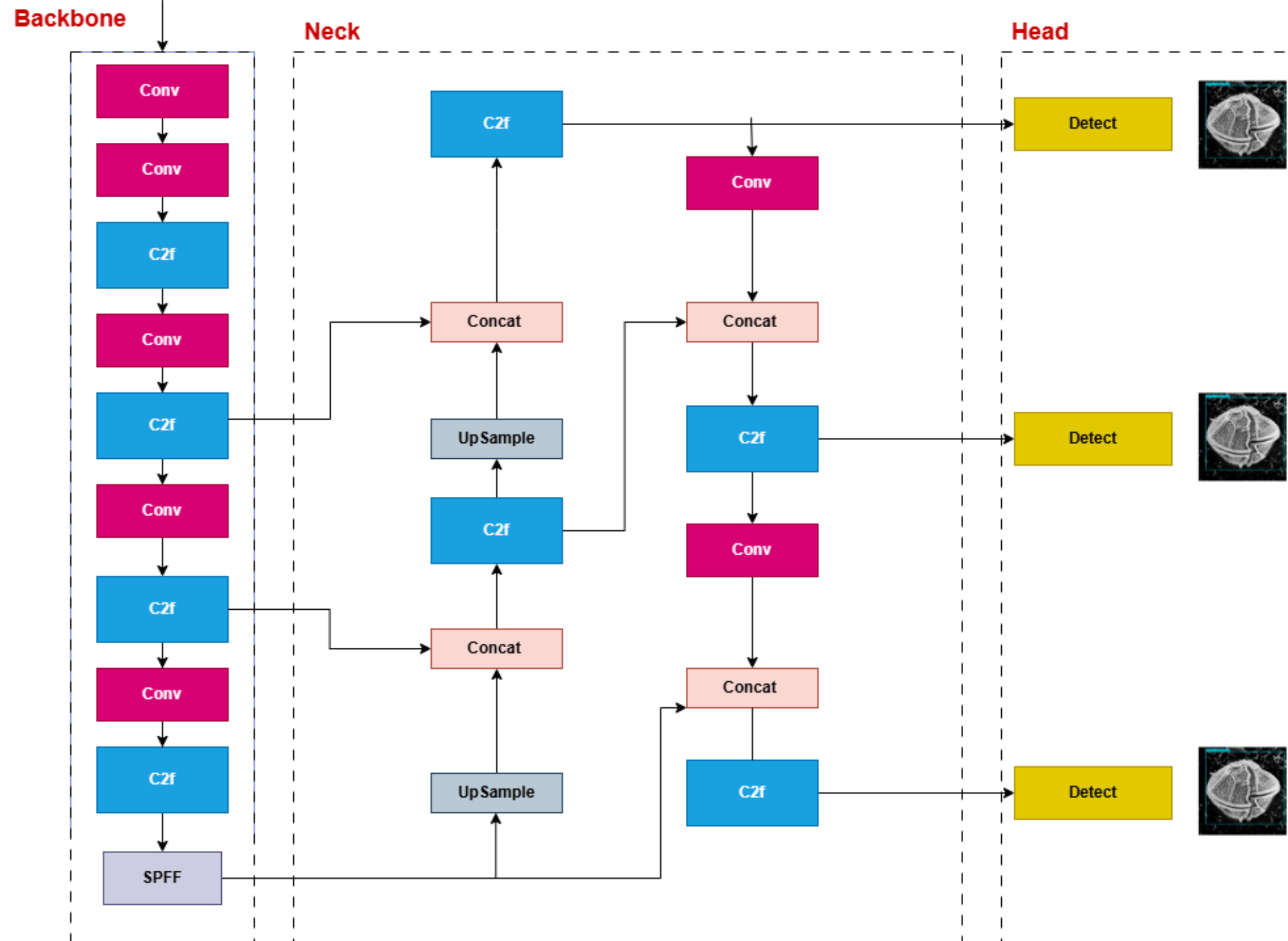
- YOLOv8: CNN-based, fast inference, high recall.
- YOLOv11: Improved feature extraction (C2f+PSA attention).
- RF-DETR: Transformer-based, precise bounding box prediction

Classification Models:

- ResNet18: Lightweight CNN for classification.
- ResNet50: Deeper variant, better accuracy but heavier.
- Applied only to *cropped cells* (pure classification).

YOLOV8'S MODEL ARCHITECTURE

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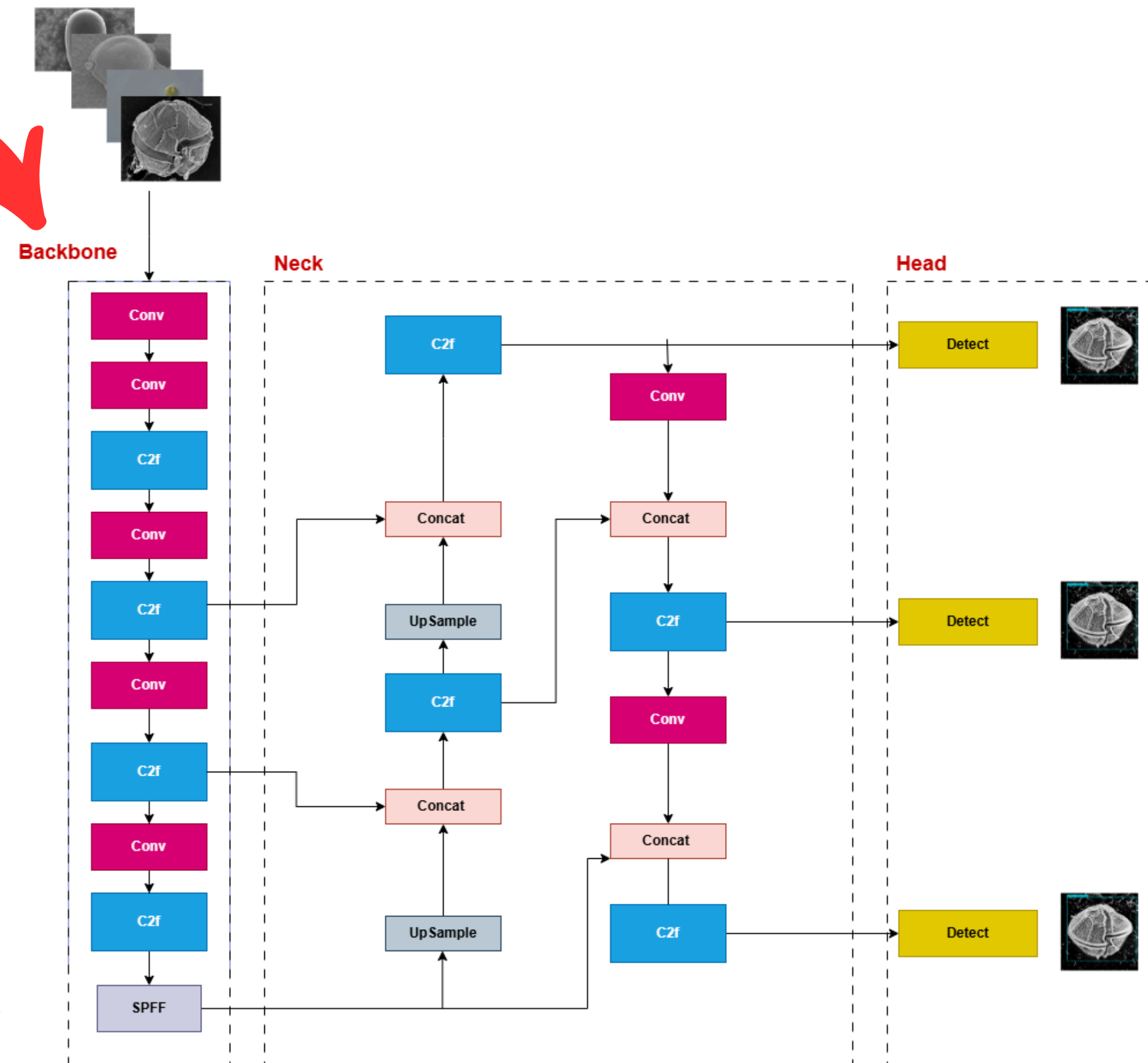
- ✓ Key Strengths:
- Supports accurate detection across varying object sizes.
 - Optimized for speed and accuracy in complex scenarios.

Backbone: Feature Extraction

Conv Layers : Capture spatial features from the input image.

C2f Modules : Efficient blocks with partial connections.

SPPF : Spatial Pyramid Pooling - Fast: Aggregates multi-scale spatial features for improved object detection.



YOLOV8'S MODEL ARCHITECTURE

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◆ Neck: Feature Fusion

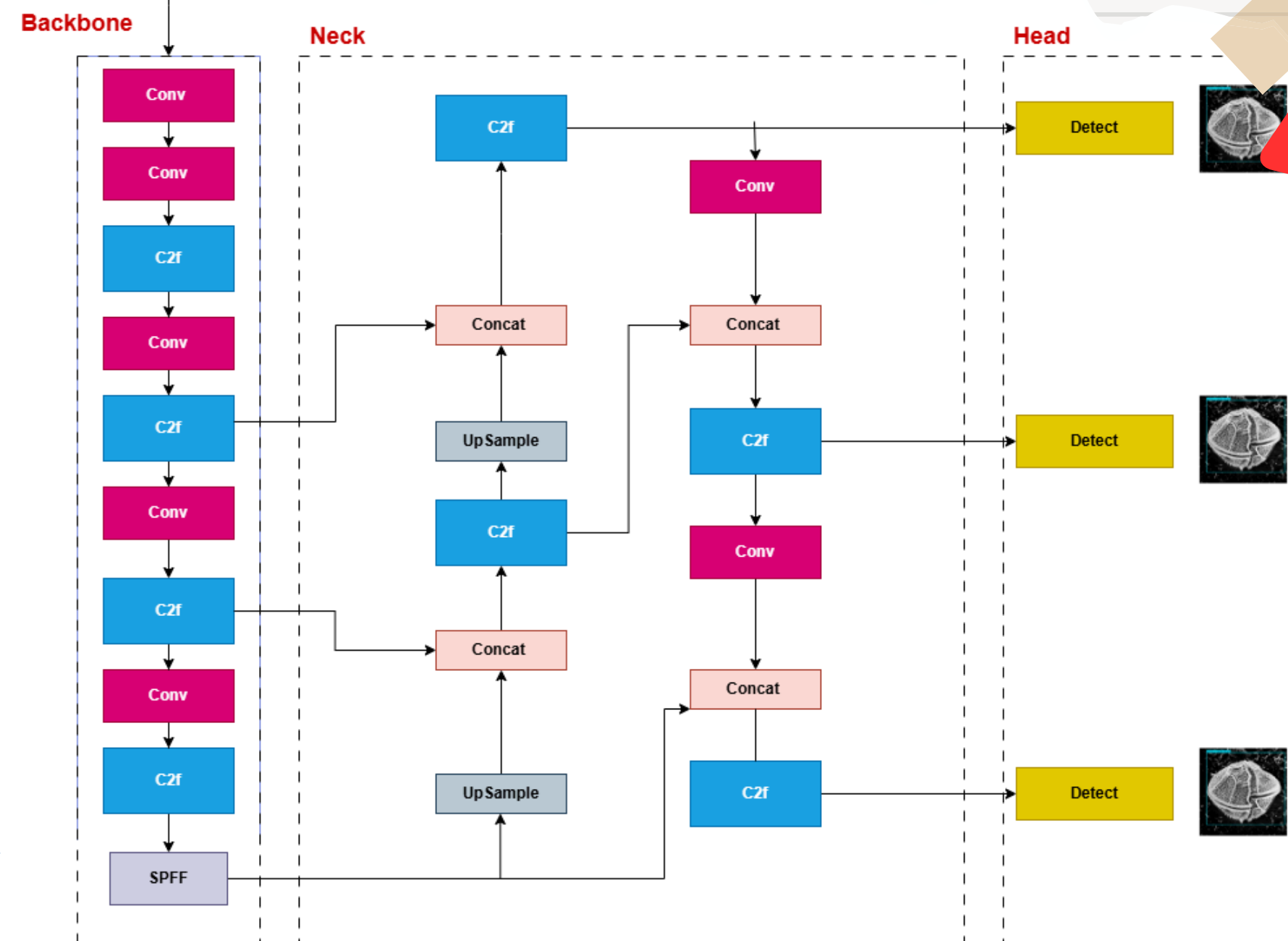
Upsample : Increases resolution to better combine features.

Concat : Merges feature maps from different layers.

C2f & Conv: Further refine features for richer representation.

→ Helps detect both small and large objects by combining shallow and deep features.

● Head: Detection Output
→ Output bounding boxes & class scores at 3 different scales.



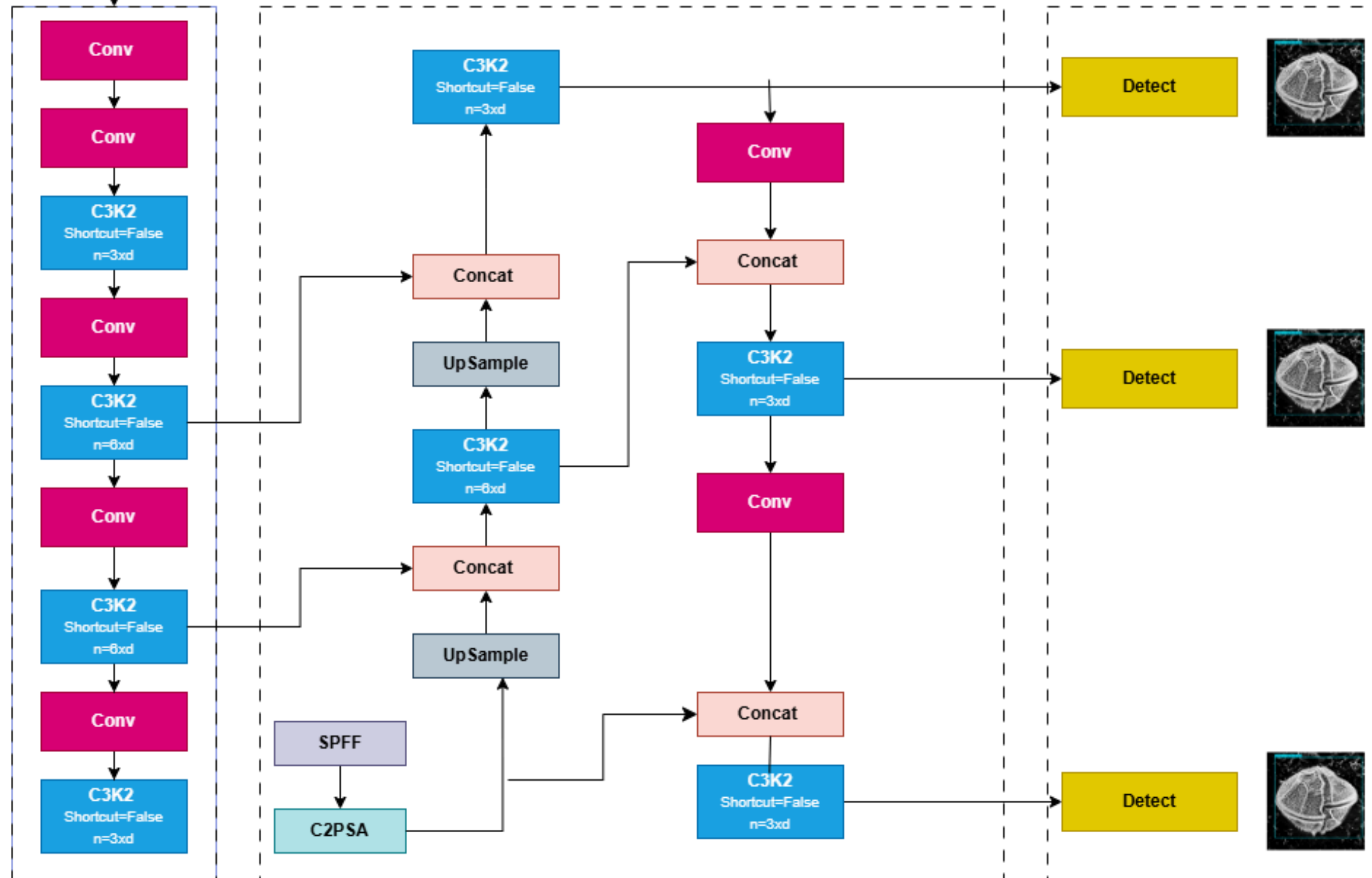
YOLOV7's MODEL ARCHITECTURE

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Backbone

Neck

Head



✓ Key Advantages:

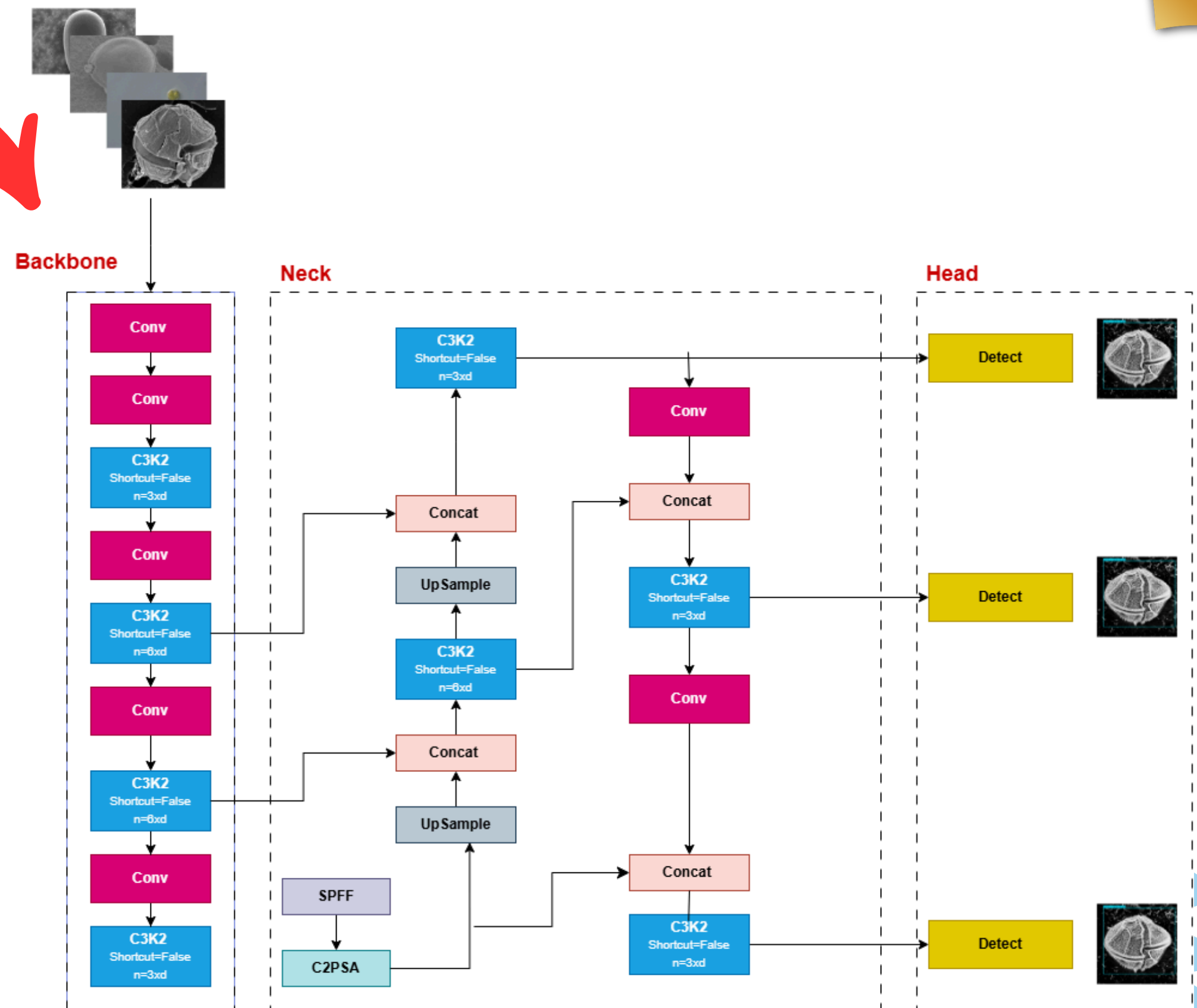
- Tailored for complex scenes with varying object sizes.
- Combines efficiency, scale awareness, and robust detection performance.

Backbone: Feature Extraction

Conv: Standard layers capturing low-level details.

C3K2 (Shortcut=False): Custom convolutional blocks (like YOLO's C3) with configurable depth (e.g., $3 \times d$, $6 \times d$)

:Progressively downsamples input while increasing feature abstraction.



YOLOV7's MODEL ARCHITECTURE

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Neck: Feature Aggregation & Fusion

Upsample: Aligns feature map resolutions across layers.

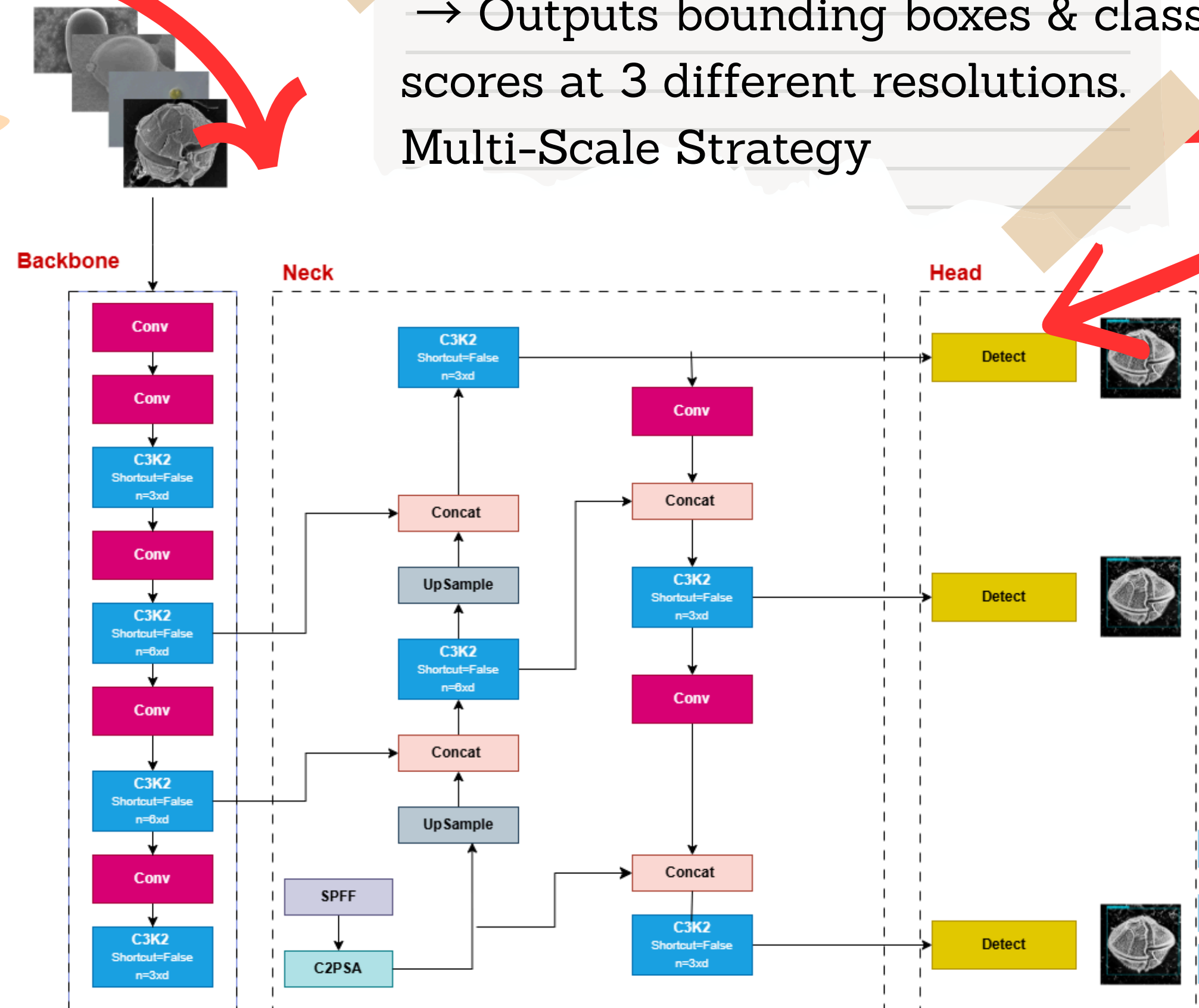
Concat: Merges features across scales (FPN/PANet style).

C3K2: Refines fused features at each scale.

SPPF (Spatial Pyramid Pooling - Fast): Expands the receptive field for better scale-invariance.

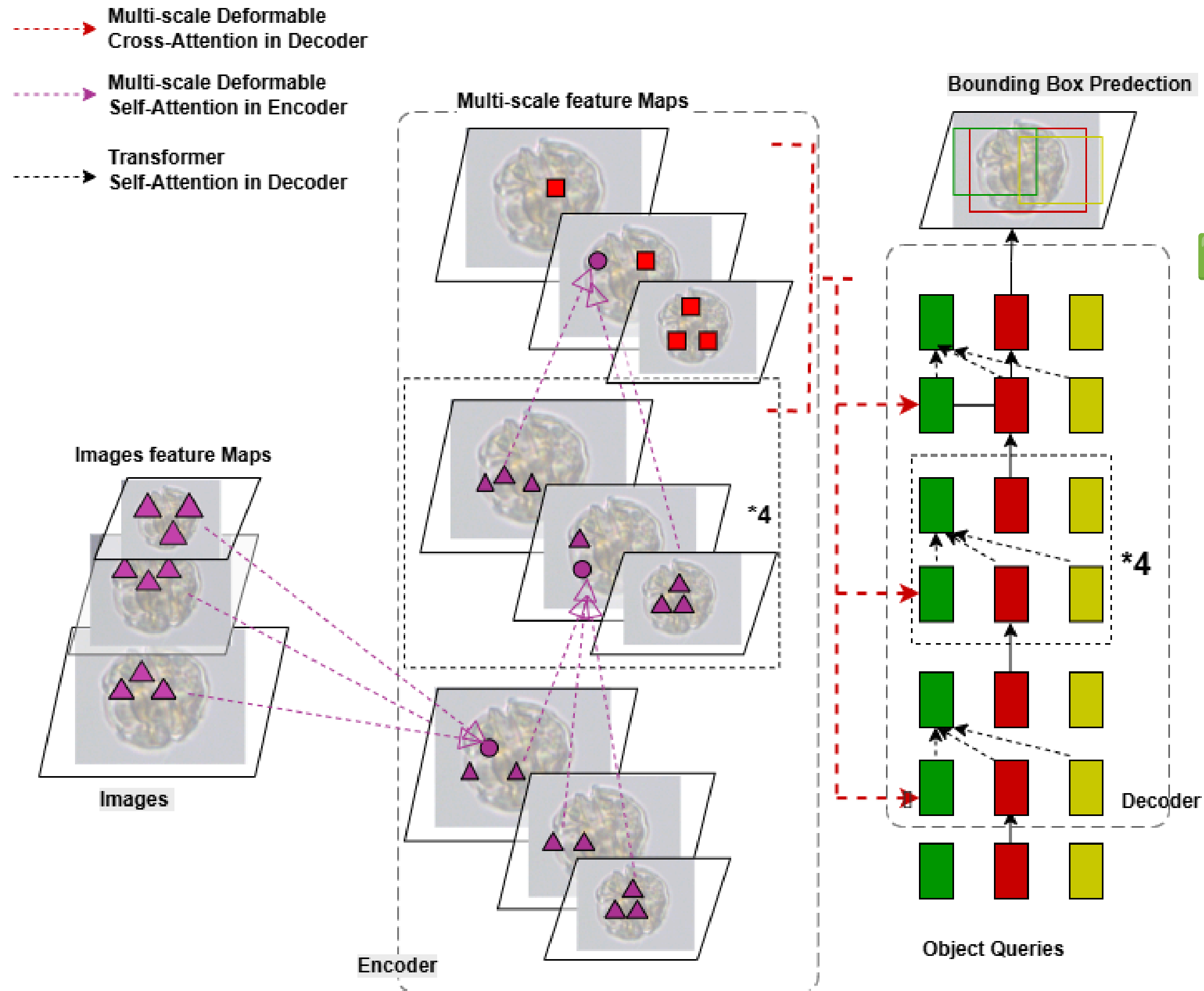
C2PSA: Custom module for advanced attention-based feature refinement.

Head: Multi-Scale Detection
→ Outputs bounding boxes & class scores at 3 different resolutions.
Multi-Scale Strategy



RF-DETR'S MODEL ARCHITECTURE

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✓ Key Strengths:

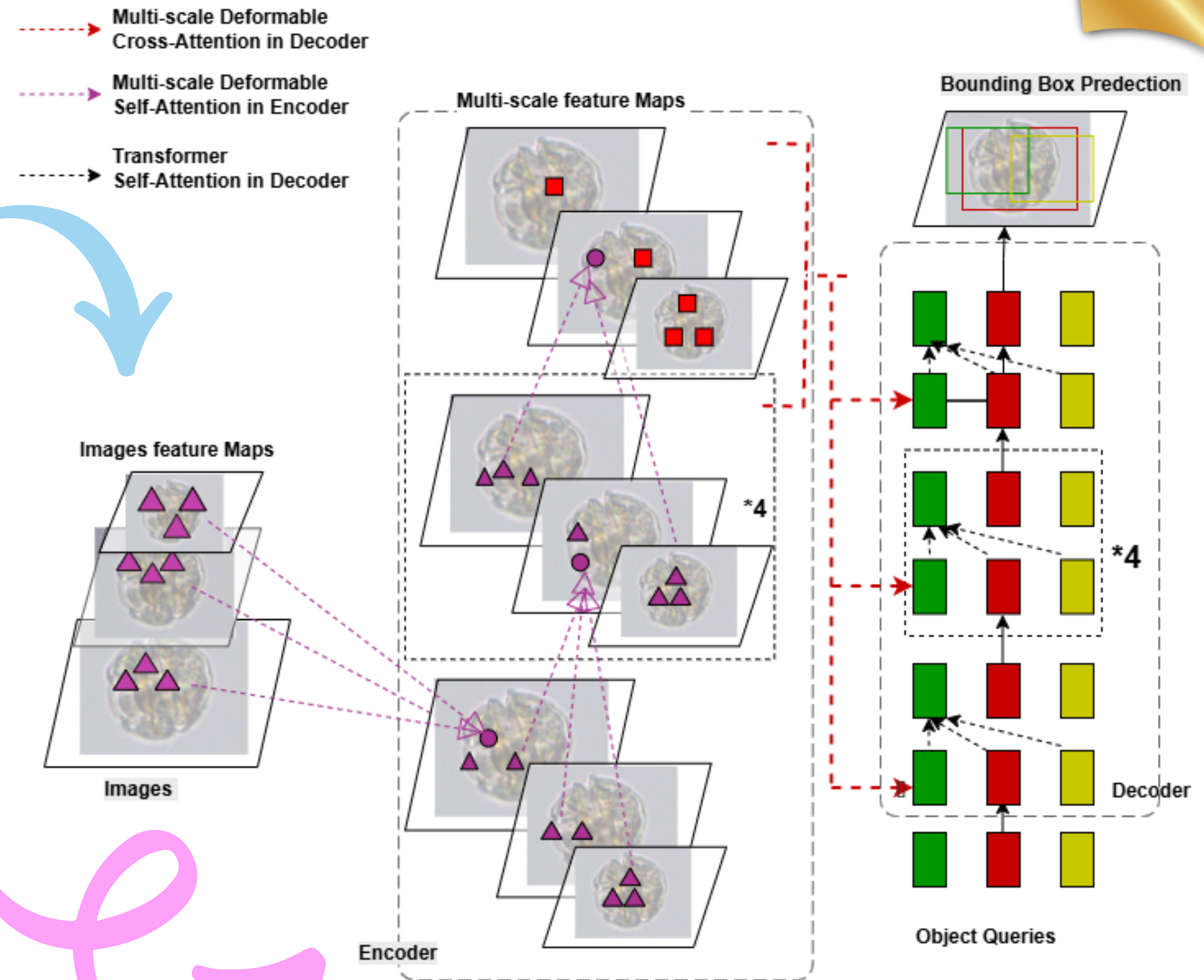
- Global reasoning across the entire image.
- Long-range dependency modeling.
- Multi-scale attention enhances performance in cluttered or complex scenes.

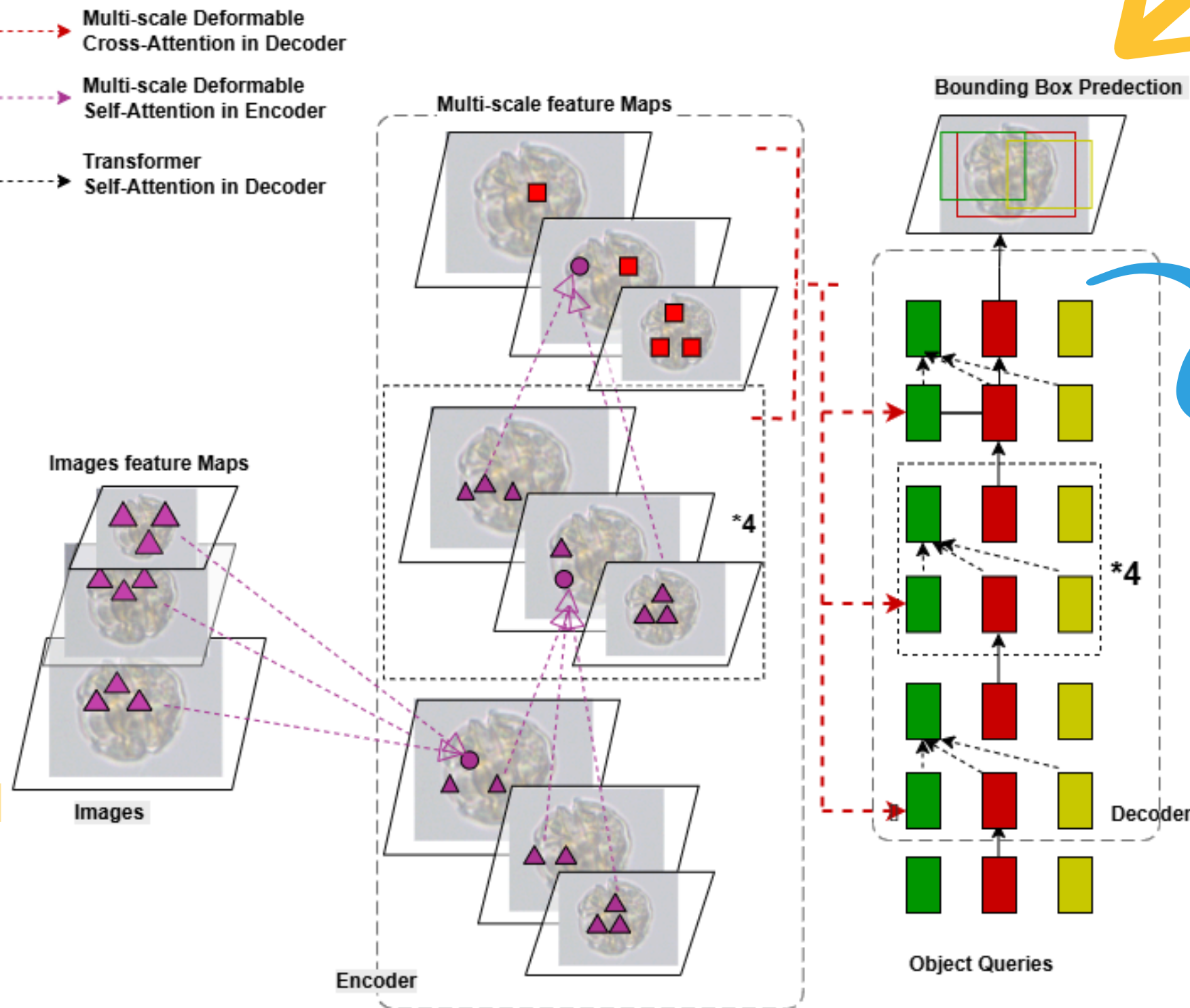
Image Feature Maps

- The input image is converted into multi-resolution feature maps.

Encoder: Adaptive Visual Encoding

- Uses multi-scale deformable self-attention.
- Dynamically focuses on important regions.
- Flexible to scale and perspective variations.





Bounding Box Prediction

- Refined queries are converted into:
 - Bounding Boxes
 - Class Labels
- Delivers accurate and robust detection results.

Decoder: Object Query Refinement

- Operates on a fixed set of object queries.

Two key mechanisms:

● Multi-Scale Deformable Cross-Attention:

Aligns queries with relevant features across resolutions.

● Self-Attention: Enables interaction among queries to capture inter-object dependencies.

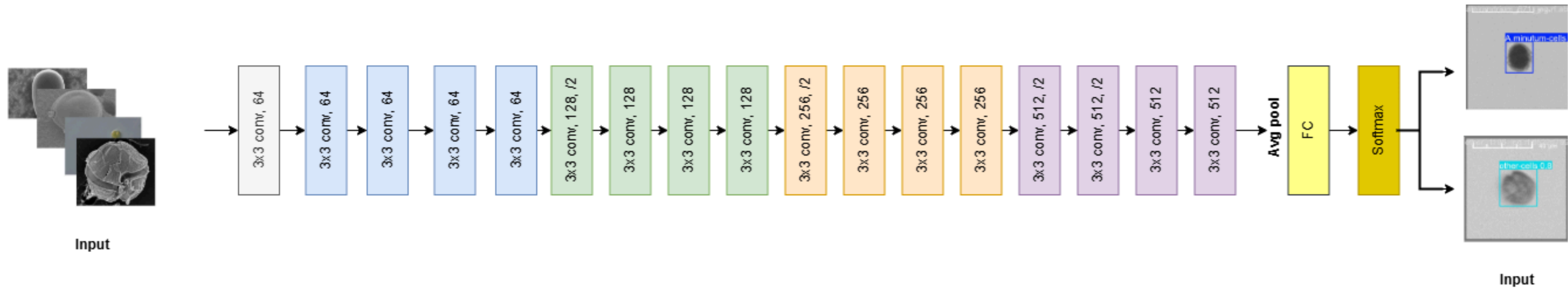
YOLOV8, YOLOV11 AND RF-DETR KEY DIFFERENCES

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Component	YOLOv8	YOLOv11	RF-DETR
Backbone	Conv layers, C2f modules, SPPF for multi-scale spatial features	Conv + C3K2 blocks (Shortcut=False), SPPF, deeper residual learning	Multi-scale deformable self-attention encoder; spatial hierarchies
Neck	Up-sample, Concat, C2f and Conv layers	Up-sample, Concat, C3K2, SPPF, C2PSA module	No traditional neck; multi-scale reasoning via attention
Head	Detect blocks at 3 scales; bounding boxes and class probabilities	Three-scale Detect head; improved multi-scale precision	Decoder with deformable cross-attention and self-attention; predicts boxes and labels
Detection Approach	CNN-based hierarchical fusion	CNN with attention modules and deeper paths	Transformer-based with global attention
Multi-scale Support	Yes – SPPF + feature fusion	Yes – Multi-scale with attention	Yes – Via deformable attention mechanisms
Strengths	Lightweight, fast, real-time detection	Balanced accuracy and depth with partial attention	Excels in cluttered scenes; strong contextual reasoning

RESNET18'S MODEL ARCHITECTURE

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✓ Key Benefits:

- Hierarchical feature extraction.
- Residual connections stabilize deep training.
- Effective generalization across complex image data.

RESNET18'S MODEL ARCHITECTURE

13

● Initial Convolution

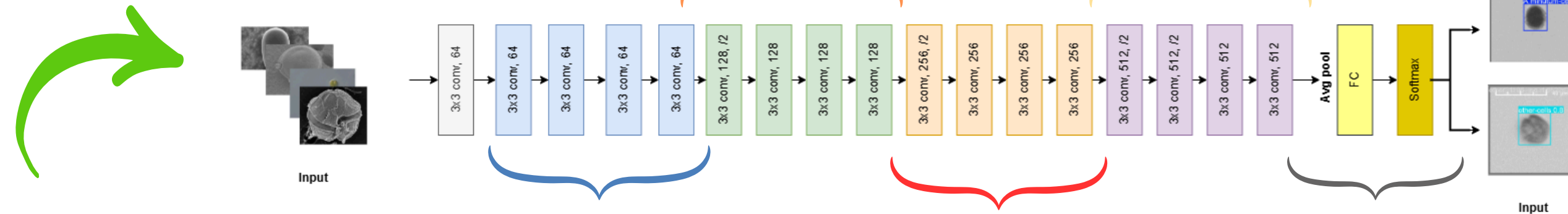
- Captures low-level features : edges, textures.

● Stage 2

- Downsampling with feature depth expansion.
- Mid-level abstraction.

● Stage 4

- Final spatial downsampling.
- Extracts high-level, class-specific features.



Output Classes:

- ◆ A.minutum-cells
- ◆ Other-cells

✚ Input Layer

- Images of plankton cells.
- Starting point for feature extraction.

● Stage 1

- Deepens early feature representations.

● Stage 3

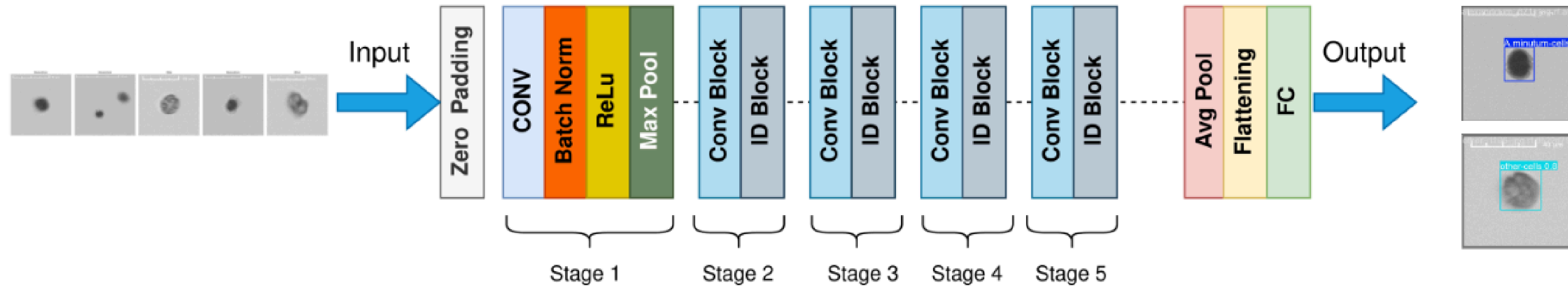
- Further spatial reduction.
- Captures complex patterns.

⚙️ Post-Convolution Layers

- Average Pooling: Reduces feature maps to single values.
- Fully Connected Layer: Outputs class scores.
- Softmax: Converts scores into class probabilities.

RESNET50'S MODEL ARCHITECTURE

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✓ Key Advantages:

- Residual Connections: Enable deep, stable training.
- Efficient Depth: Balances computational cost and representational power.
- High Precision: Effectively separates Alexandrium cells from other plankton types.
- Complements Detection Models: Strengthens overall analytical accuracy.

RESNET50'S MODEL ARCHITECTURE

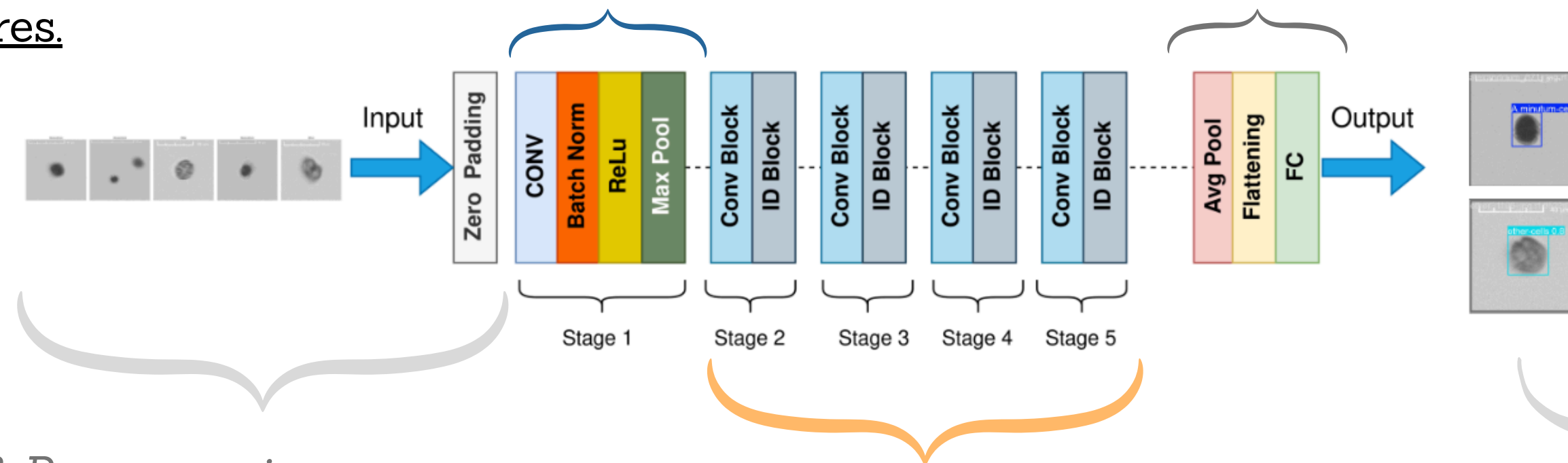
13

◆ Stage 1: Initial Feature Extraction

- **Convolutional Layer:** Captures basic visual features (edges, textures).
- **Batch Normalization:** Stabilizes and accelerates training.
- **ReLU Activation:** Enables learning of complex patterns.
- **Max Pooling:** Downsamples feature maps, preserving key features.

⚙️ Post-Convolution Layers

- **Average Pooling:** Summarizes each feature map into a single value.
- **Flattening:** Converts 2D maps into a 1D feature vector.
- **Fully Connected (FC) Layer:** Performs final classification.



🖼️ Input & Preprocessing

- Input Layer: images of plankton cells.
- Zero Padding: Maintains spatial dimensions during convolutions.

◆ Stages 2–5: Residual Learning:

1. Enable skip connections for smooth gradient flow.
2. Support deep network training without vanishing gradients.
3. Extract increasingly abstract and hierarchical features.

📋 Output Classes:

- ◆ A.minutum-cells
- ◆ Other-cells

RESNET18 AND RESNET50 KEY DIFFERENCES

13

Component	ResNet18	ResNet50
Depth	18 layers	50 layers
Building Block	Basic-Block (2 layers)	Bottleneck Block (3 layers)
Conv1	7x7 conv, 64 filters, stride 2	7x7 conv, 64 filters, stride 2
Stage 1	2 x Basic-Block (64 filters)	3 x Bottleneck (256 filters)
Stage 2	2 x Basic-Block (128 filters)	4 x Bottleneck (512 filters)
Stage 3	2 x Basic-Block (256 filters)	6 x Bottleneck (1024 filters)
Stage 4	2 x Basic-Block (512 filters)	3 x Bottleneck (2048 filters)
Strengths	Lightweight, fast inference	Deeper features, better accuracy

HYPERPARAMETERS FINE-TUNING

Hyperparameters Settings

✦ *Optimizer*

Value : AdamW

✦ *Batch-size*

Value : 16

✦ *Learning Rate*

Value : lr-4e
around 0.00167

✦ *Number of Epochs*

Value : 100

```
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining
best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.001667, momentum=0.9) with parameter groups 77 weight(decay=0.0), 84 w
eight(decay=0.0005), 83 bias(decay=0.0)
Image sizes 960 train, 960 val
Using 4 dataloader workers
Logging results to runs/detect/alexandrium_yolov8
Starting training for 100 epochs...
```



The learning rate and optimizer were set automatically by the models as we set the model to auto so that it chooses what would work best with the train data.

EVALUATION METRICS

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Object Detection Evaluation :

Metrics used:

mAP@0.50 – Mean Average Precision at 50% IoU (Intersection over Union)

mAP@0.50:0.95 – Averaged over different thresholds

Precision – Correct detections out of all positive predictions

Recall – Correct detections out of all actual objects

Why these metrics?

Detection is not just about finding the right object—it's about finding the right object in the right location with high confidence.

EVALUATION METRICS

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Classification Evaluation :

Metrics used:

Accuracy – Correct predictions / total predictions (confusion matrix would give us a better understanding across different classes)

F1-Score – Balance between precision and recall (important for class imbalance)

Reason: We had two classes (A. minutum vs other cells), and correct classification is key for ecological monitoring.

EVALUATION METRICS

Confusion Matrix

TP (True Positive)

The model correctly predicts a positive class.

FN (False Negative)

The model incorrectly predicts a negative class.

FP (False Positive)

The model incorrectly predicts a positive class.

TN (True Negative)

The model correctly predicts a negative class.

TP True Positives	FN False Negatives Type II error
FP False Positives Type I error	TN True Negatives

RESULTS FOR THE FIRST DATASET

Detection results

Model	mAP@0.50	mAP@0.50:0.95	Conclusion
YOLOv8	0.89	0.63	High object detection performance, but less accurate localization.
YOLOv11	0.87	0.62	Similar to YOLOv8, suitable for fast detection with moderate accuracy.
RF-DETR	0.80	0.73	Best for accurate localization and spatial precision, despite lower mAP@0.50.
<u>Conclusion:</u> Use YOLO for speed, RF-DETR for accuracy.			

RESULTS FOR THE FIRST DATASET

Classification results

Model	Accuracy	Precision	Recall	Conclusion
YOLOv8	0.64	0.71	0.89	High recall; detects most instances but with more false positives.
YOLOv11	0.65	0.97	0.67	High precision; fewer false positives but may miss many true cases.
RF-DETR	0.83	0.83	0.87	Balanced performance; best overall for small or imbalanced datasets.
Conclusion RF-DETR is most reliable for classification.				

RESULTS FOR THE FIRST DATASET

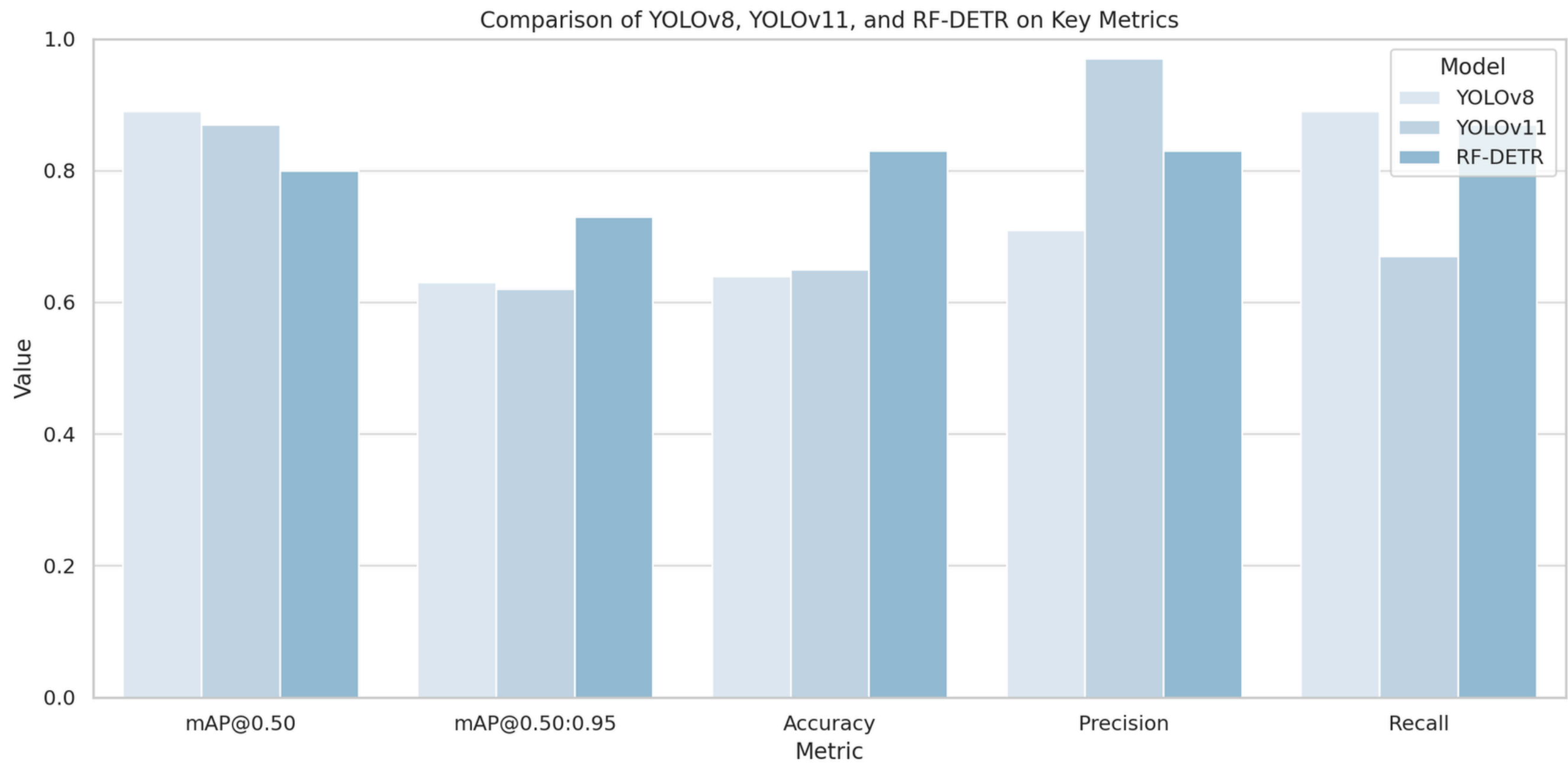


Figure 1: Bar Chart Comparing Detection and Classification Results on the first Dataset

RESULTS FOR THE FIRST DATASET

Detection and Classification results discussion

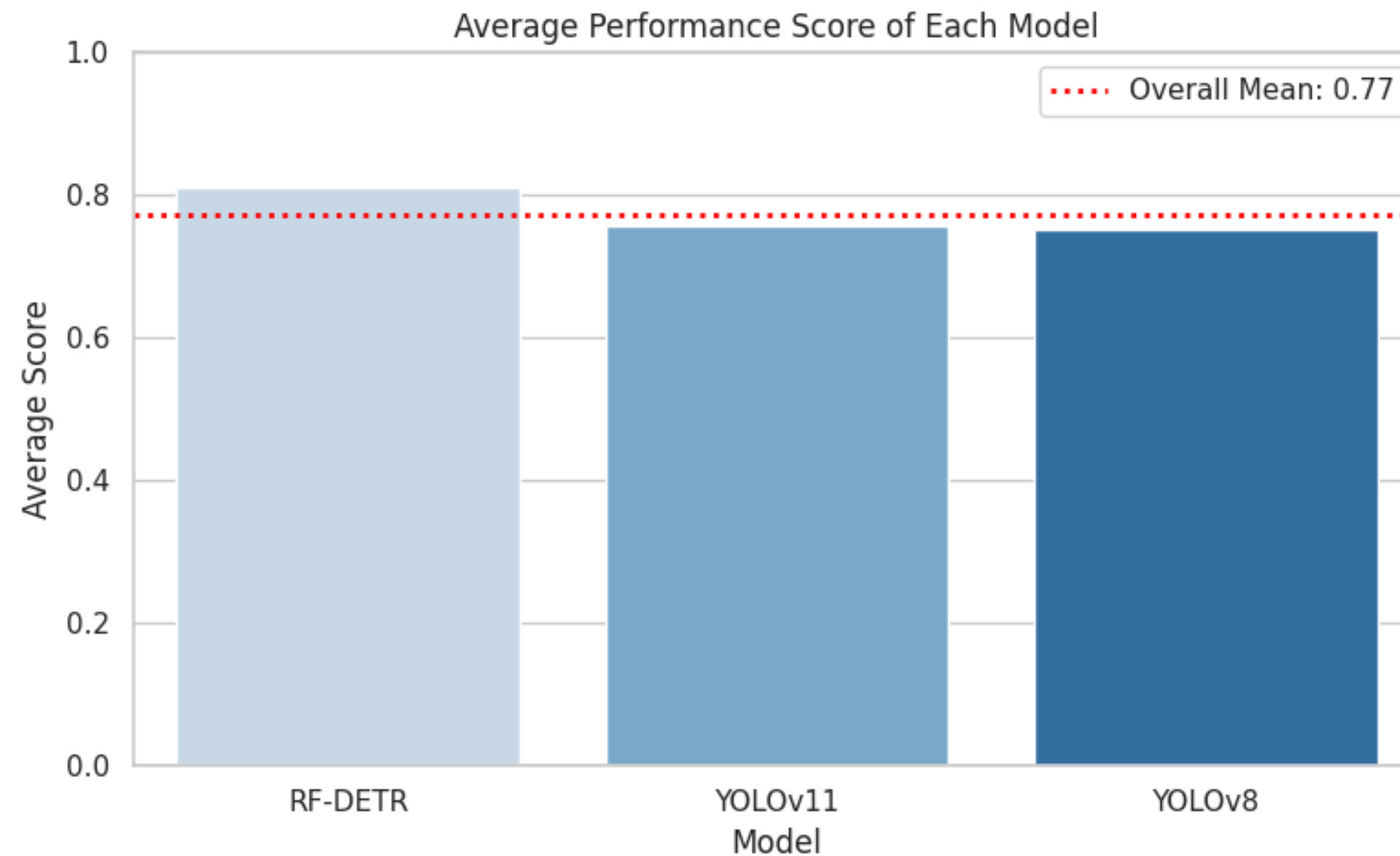
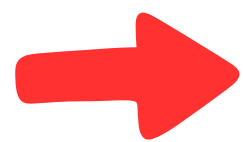


Figure 1: Overall Performance Results on the first Dataset

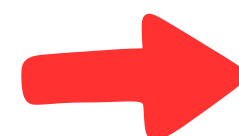


Since RF-DETR delivered the *best performance*, we chose to apply it to our second dataset as well.

EXPIREMENTATION PHASES

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Phase	Description	Problem Identified	Results
Phase 1: Initial Setup	Used small versions of YOLOv8 and YOLOv11 with 70/20/10 data split.	Poor accuracy and inconsistent detection performance.	YOLOv8S: 83%, YOLOv11S: 80%
Phase 2: Data Augmentation & Label Verification	Applied augmentation techniques , manually corrected labels and change images sizes till we had seven version .	No significant improvement despite better data quality.	No major performance gain observed.
Phase 3: Pipeline & Training Strategy Modification	Adjusted input resolution, froze initial layers, resumed training from checkpoint, extended training with all layers and change the models into medium version .	Models struggled with low-res microscopic images.	YOLOv8M: 85%, YOLOv11M: 81%
Phase 4: Dataset Re-balancing	Identified class imbalance and redistributed samples for equal representation.	Large disparity between class distributions.	Balanced dataset created with 80/10/10 split.



Dataset number two is **ready** for training !!!

RESULTS FOR THE SECOND DATASET

RF-DETR results:

Phase	Description	Problem Identified	Outcome
Phase 1: Initial Setup	Trained model with: • 50 epochs, • batch size = 2, • learning rate = 1e-4.	Despite promising results, performance was limited by small batch size and under-optimized hyperparameters.	mAP@50 = 0.91, mAP@75 = 0.74, mAP@50:95 = 0.62
Phase 2: Hyperparameter Tuning Attempt	Increased batch size to 4	Technical limitation: Inadequate GPU memory and compute prevented training with higher batch size.	Training could not proceed as planned; model evaluation halted due to hardware constraints.

Phase	Description	Outcome	Problem Identified
Phase 1: Initial Setup	Trained model with: 50 epochs, batch size = 2, learning rate = 1e-4.	mAP@50 = 0.91, mAP@75=0.74, mAP@50:95 = 0.62	Despite promising results, performance was limited by small batch size and under-optimized hyperparameters.
Phase 2: Hyperparameter Tuning Attempt	Increased batch size to 4	Training could not proceed as planned; model evaluation halted due to hardware constraints.	Technical limitation: Inadequate GPU memory and compute prevented training with higher batch size.

HIGHLIGHTED RESULTS

✦ Summary of the results obtained by the three proposed models:

	Precision	mAP@0.5	mAP@0.5:0.95	Inference Time
YOLOv8	0.869	0.862	0.541	~21 ms
YOLOv11	0.894	0.879	0.558	~25 ms
RF-detr	0.812	0.827	0.502	~54 ms

Conclusion: YOLOv11 gave the *highest* precision and mAP, making it the best detector for our task.

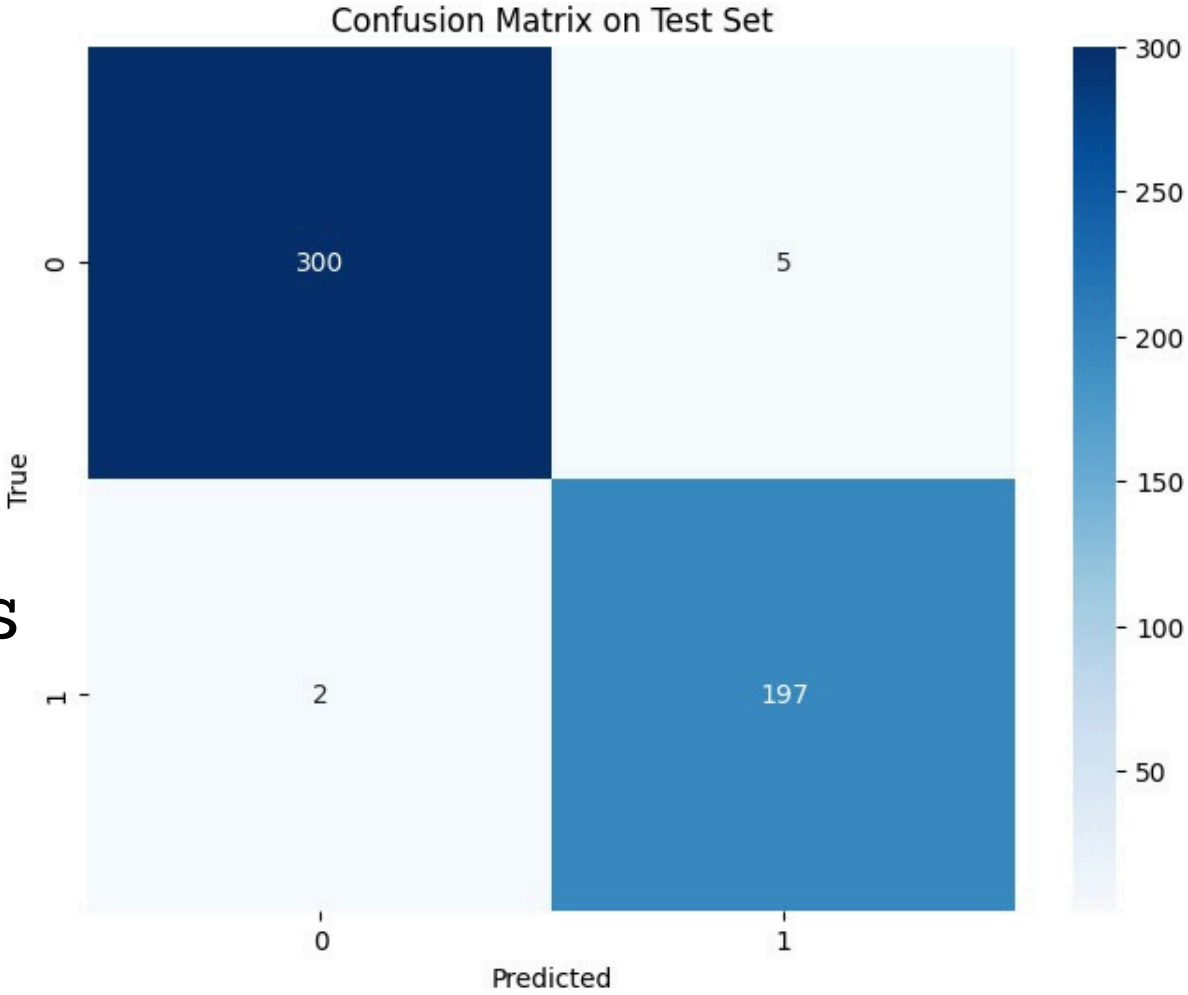
ResNet Classification Results (on Cropped Cell Images):

Summary of the results obtained by the three proposed models:

	Accuracy	F1-Score	Notes
ResNet18	0.994	0.99	Best classifier overall ✓
ResNet50	0.987	0.98	Slightly lower performance

✓ Confusion Matrix (ResNet18):

A. minutum → Correctly classified: 99.4%
Other plankton → Also high accuracy, few false positives



RESULTS FOR THE SECOND DATASET

YOLO models results:

Classification results:

Model	Accuracy	Precision	Recall	Conclusion
YOLOv8	90.5%	0.87	0.87	Strong balance , good overall performance.
YOLOv11	93.0%	0.89	0.86	Higher accuracy and precision, slightly lower recall.
ResNet18	99.4%	0.992	0.9937	Highest performance across all metrics.
ResNet50	98.1%	0.9849	0.9828	Excellent results, just below ResNet18.
Conclusion: ResNet18 is the most accurate and reliable classifier.				

ResNet Classification Results (on Cropped Cell Images):

✦ Summary of the results obtained by the three proposed models:

	Accuracy	F1-Score	Notes
			✓
ResNet18	0.994	0.99	Best classifier overall
ResNet50	0.987	0.98	Slightly lower performance

RESULTS FOR THE SECOND DATASET

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Highlighted Results (Second Dataset):

YOLOv11 Precision: 0.89

YOLOv11 mAP@0.50: ~0.88

ResNet18 Accuracy: 0.994

ResNet18 F1-Score: ~0.99

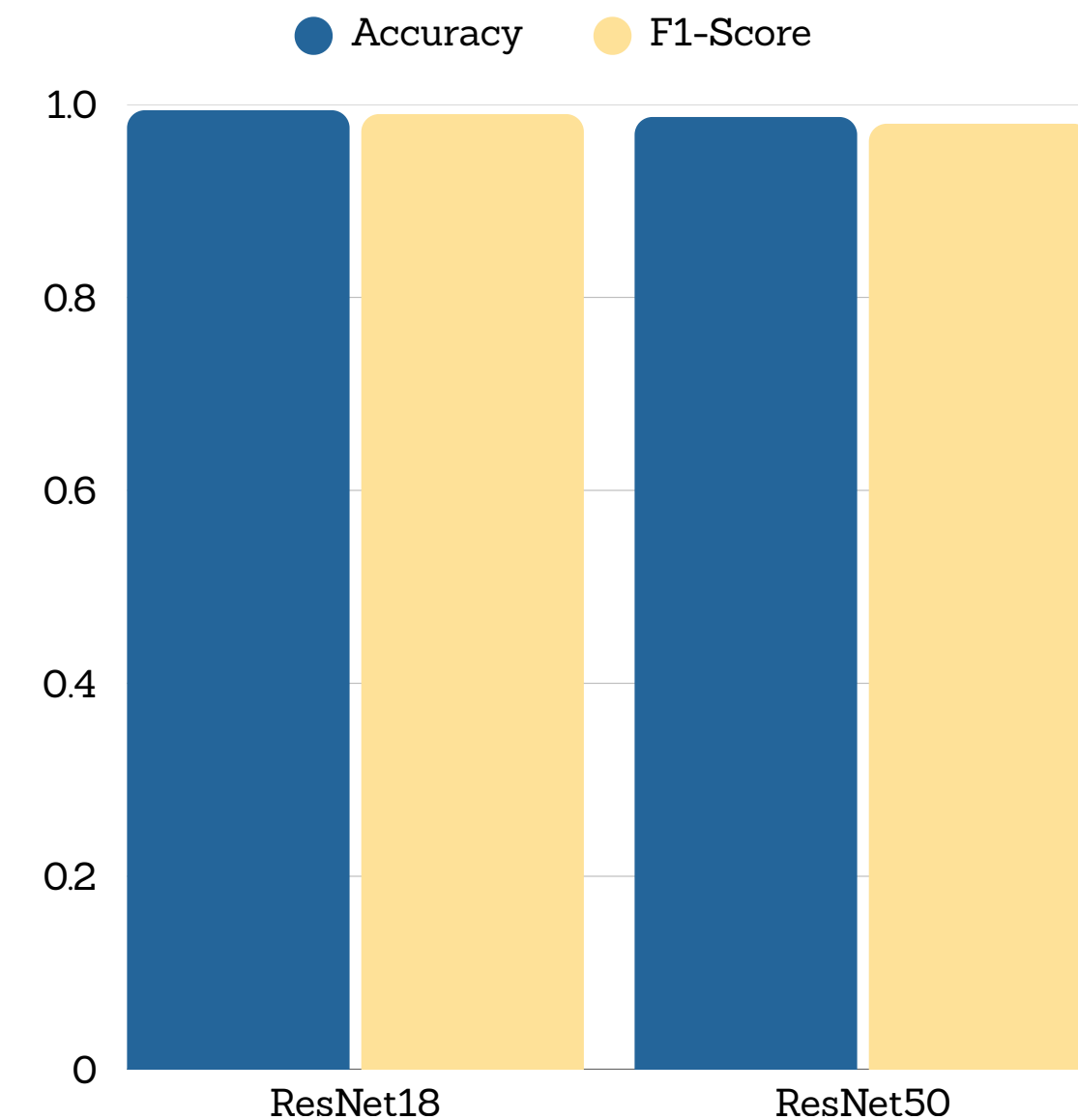


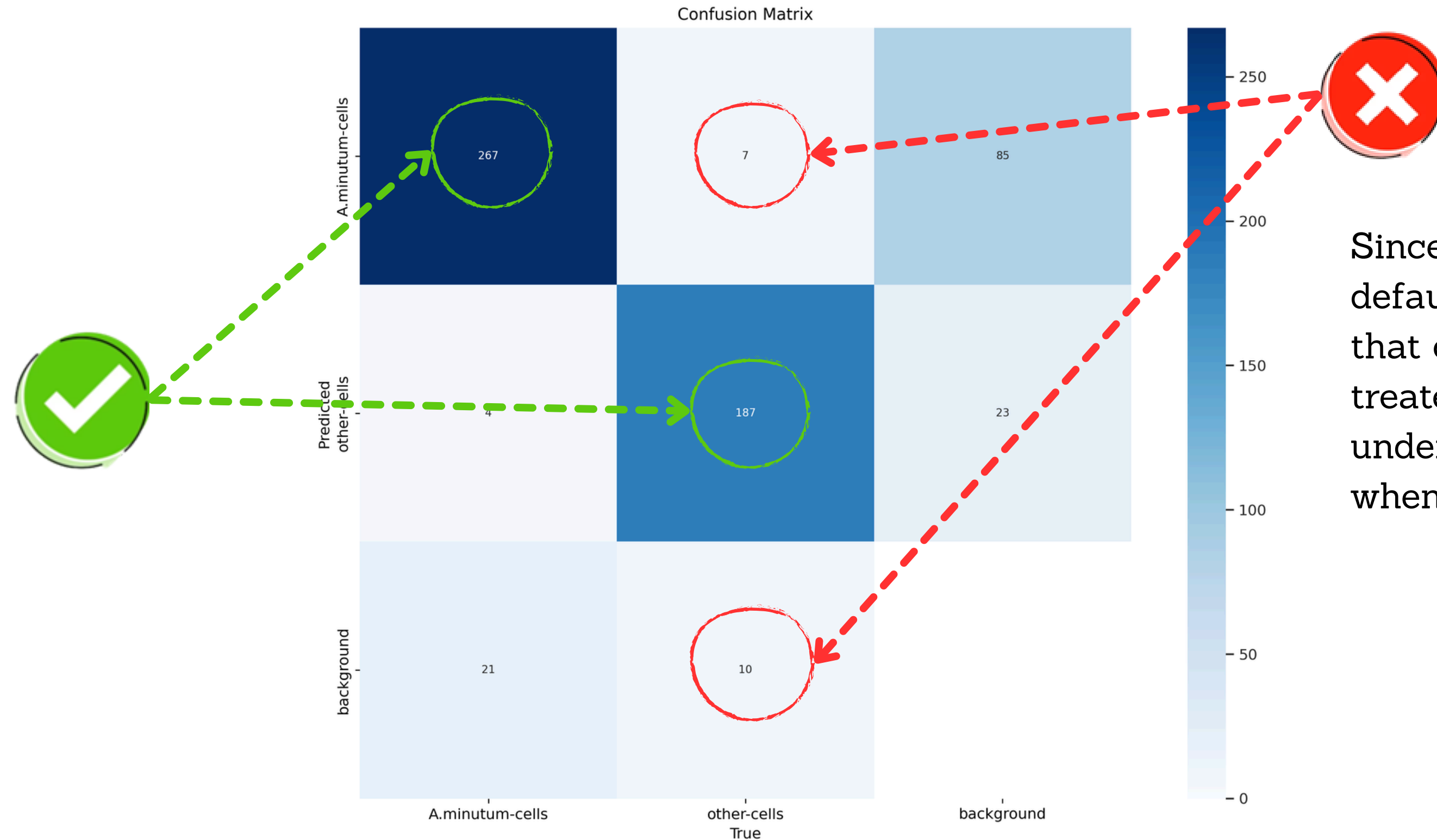
Figure 2: Bar chart Comparing Classifier Performance

Figure 1: Bar Chart Comparing Detection Results

RESULTS FOR THE SECOND DATASET

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- Confusion matrix obtained by the YOLOv11m model:



HIGHLIGHTED RESULTS

ResNet Classification Results (on Cropped Cell Images):

Resnet18 Classification Report:

Test Classification Report:				
	precision	recall	f1-score	support
0	0.9934	0.9836	0.9885	305
1	0.9752	0.9899	0.9825	199
accuracy			0.9861	504
macro avg	0.9843	0.9868	0.9855	504
weighted avg	0.9862	0.9861	0.9861	504

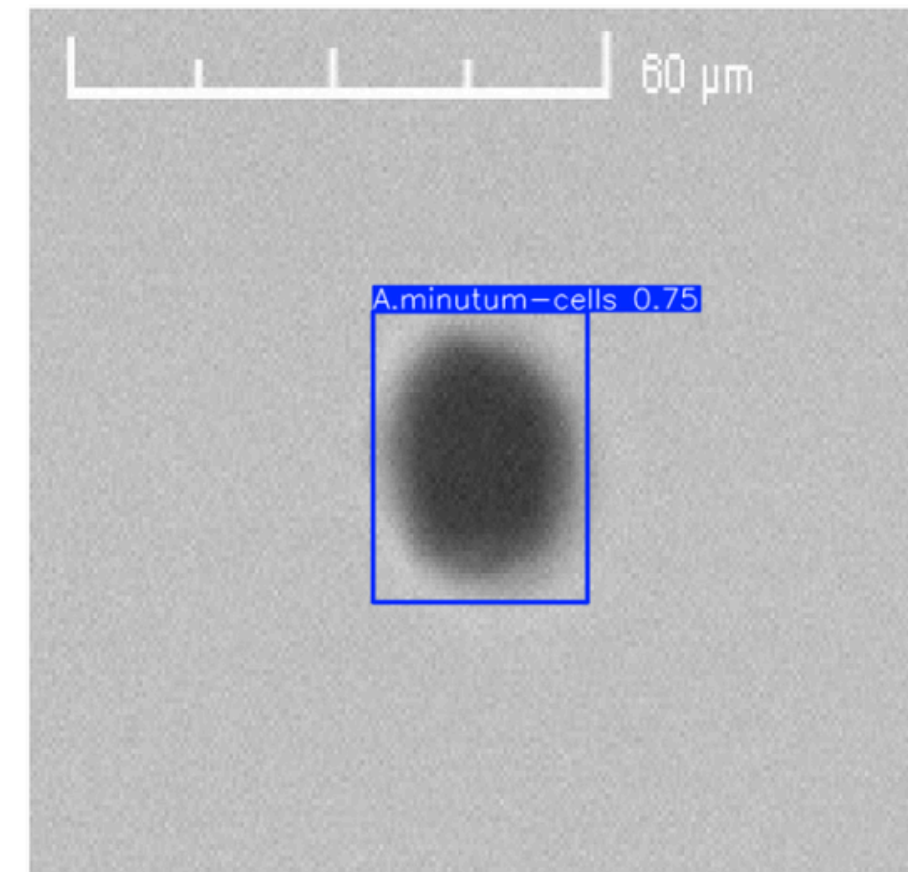
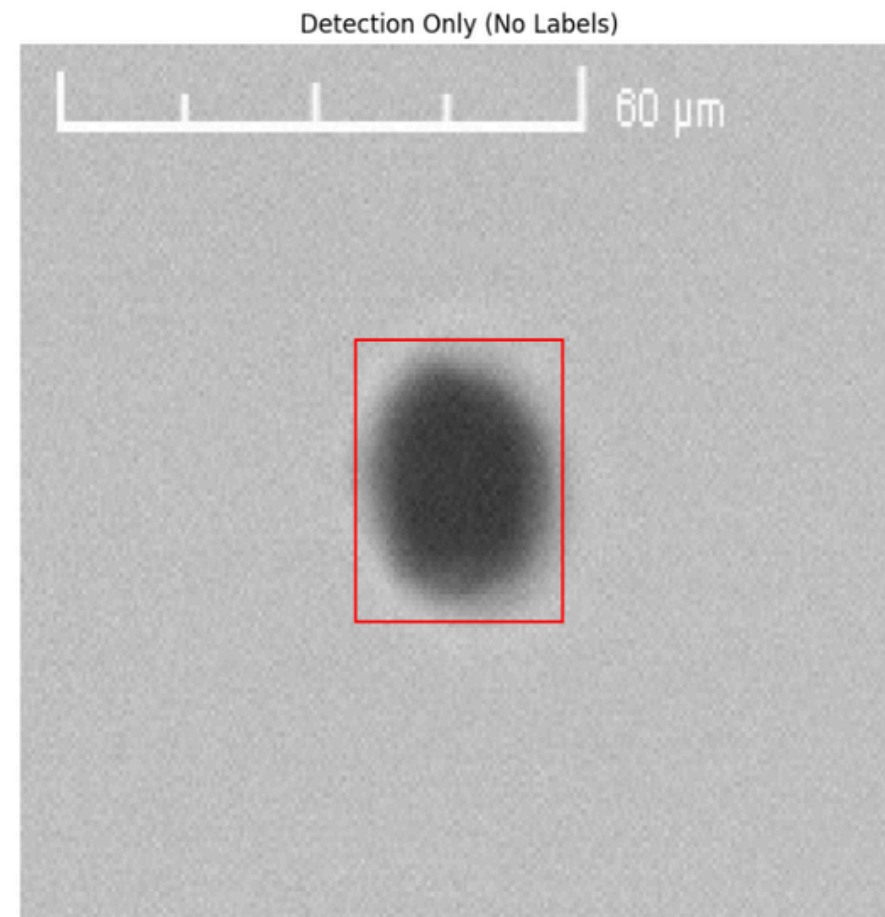
Resnet50 Classification Report:

Test Classification Report:				
	precision	recall	f1-score	support
0	0.9806	0.9967	0.9886	305
1	0.9948	0.9698	0.9822	199
accuracy			0.9861	504
macro avg	0.9877	0.9833	0.9854	504
weighted avg	0.9863	0.9861	0.9861	504

HIGHLIGHTED RESULTS



Before vs After Detection



🎯 Final Summary

Best Pipeline = YOLOv11 + ResNet18 ✓

Highest detection precision

Best classification accuracy

Fast enough for real-time deployment

FINAL PIPELINE SUMMARY

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- End-to-end pipeline for detecting and classifying *A. minutum*
- Robust to variations in scale, lighting, and cell type
- Supports real-time ecological monitoring

CONCLUSION

- ◆ Developed an AI-based solution for *A. minutum* detection

- ◆ YOLOv11 + ResNet18 yielded the *best performance*

- ◆ Contributes to early warning systems for HABs.
- ◆ Can be scaled for broader marine bio-surveillance.

FUTURE WORK

- ◆ Include more species and mixed blooms.
- ◆ Deploy on marine research stations in real-time.
- ◆ Improve transformer-based models (like RF-DETR).
- ◆ Collaborate with environmental monitoring institutions.



THANK YOU FOR
YOUR TIME AND
ATTENTION

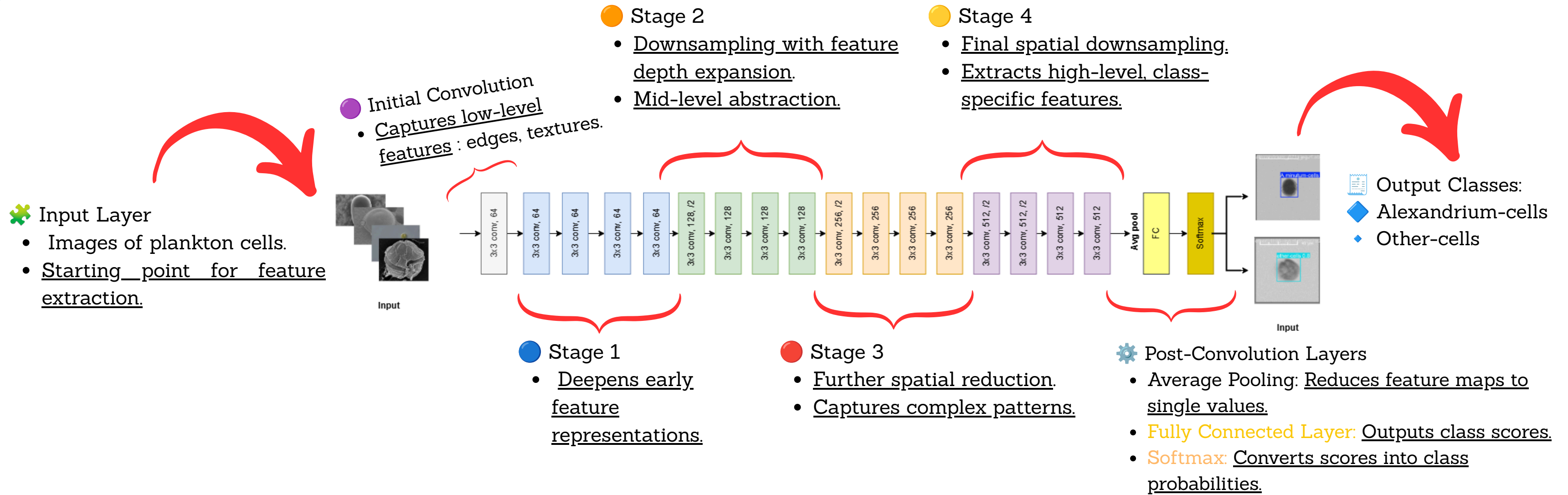
STATE OF ART COMPARAISON

Exactitude	
ResNet50-DecisionTree [1]	73.80%
ResNet50-GaussianNB [1]	65.50%
ResNet50 avec apprentissage par transfert	74.11%

Model Comparison

- YOLOv8: Faster, higher recall
- YOLOv11: Slightly slower but better precision
- RF-DETR: Good with cluttered scenes but needs refinement
- ResNet18 > ResNet50 for your use-case (lighter, more accurate)

RESNET18'S MODEL ARCHITECTURE



- ✓ **Key Benefits**
- Hierarchical feature extraction.
 - Residual connections stabilize deep training.
 - Effective generalization across complex image data.

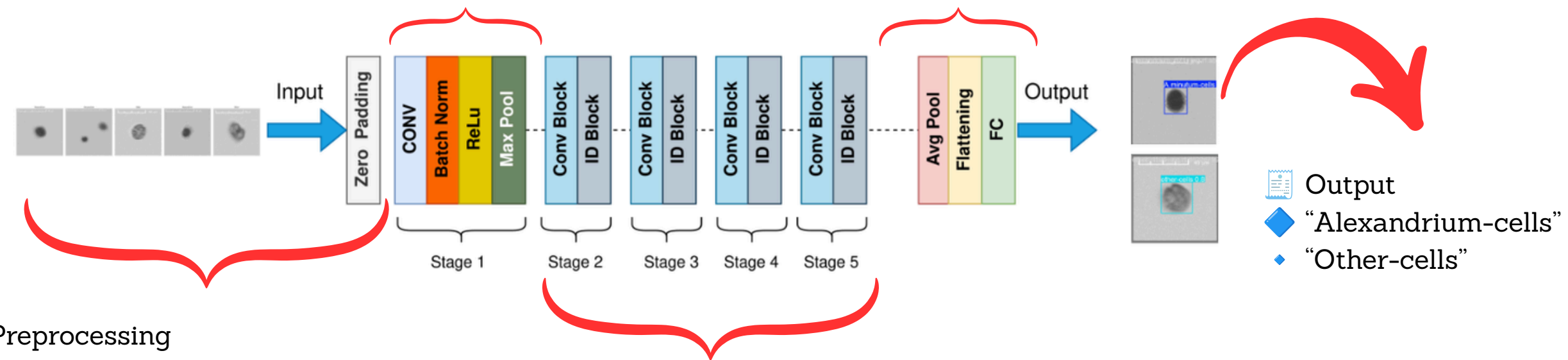
RESNET50'S MODEL ARCHITECTURE

◆ Stage 1: Initial Feature Extraction

- **Convolutional Layer:** Captures basic visual features (edges, textures).
- **Batch Normalization:** Stabilizes and accelerates training.
- **ReLU Activation:** Enables learning of complex patterns.
- **Max Pooling:** Downsamples feature maps, preserving key features.

⚙️ Post-Convolution Layers

- **Average Pooling:** Summarizes each feature map into a single value.
- **Flattening:** Converts 2D maps into a 1D feature vector.
- **Fully Connected (FC) Layer:** Performs final classification.



🖼️ Input & Preprocessing

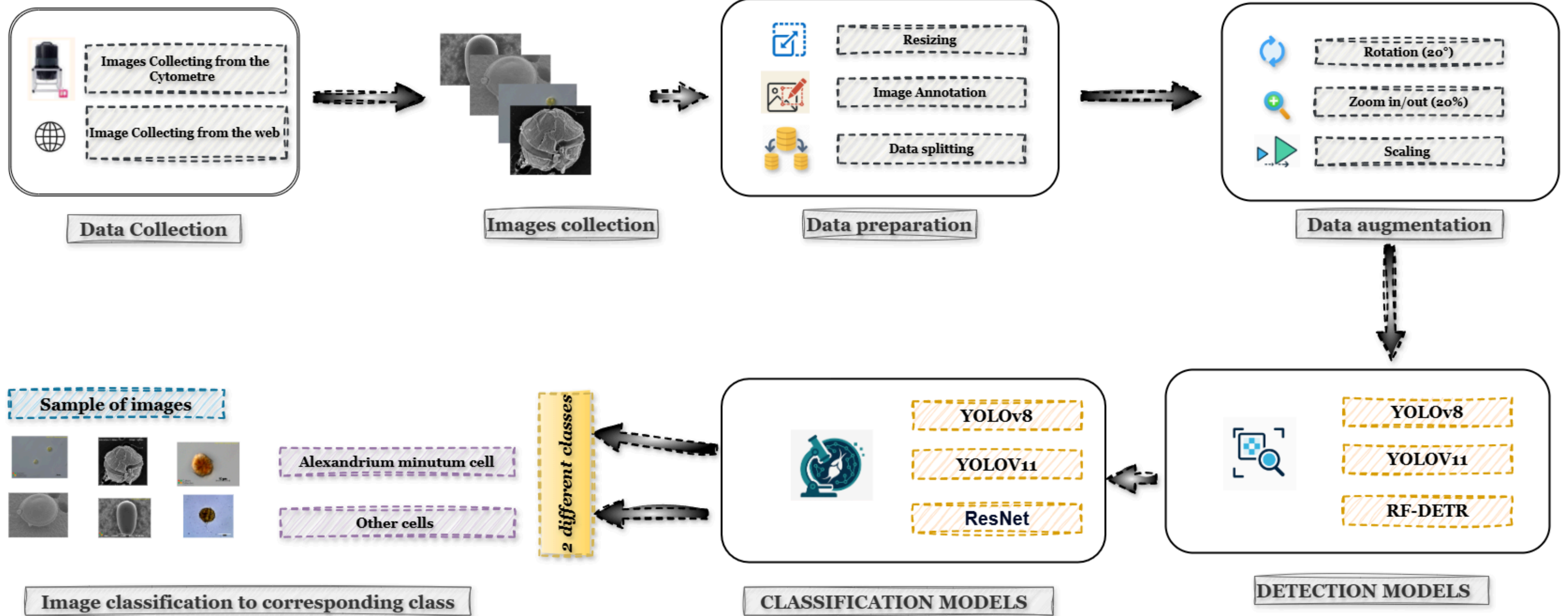
- Input Layer: images of plankton cells.
- Zero Padding: Maintains spatial dimensions during convolutions.

◆ Stages 2–5: Residual Learning:

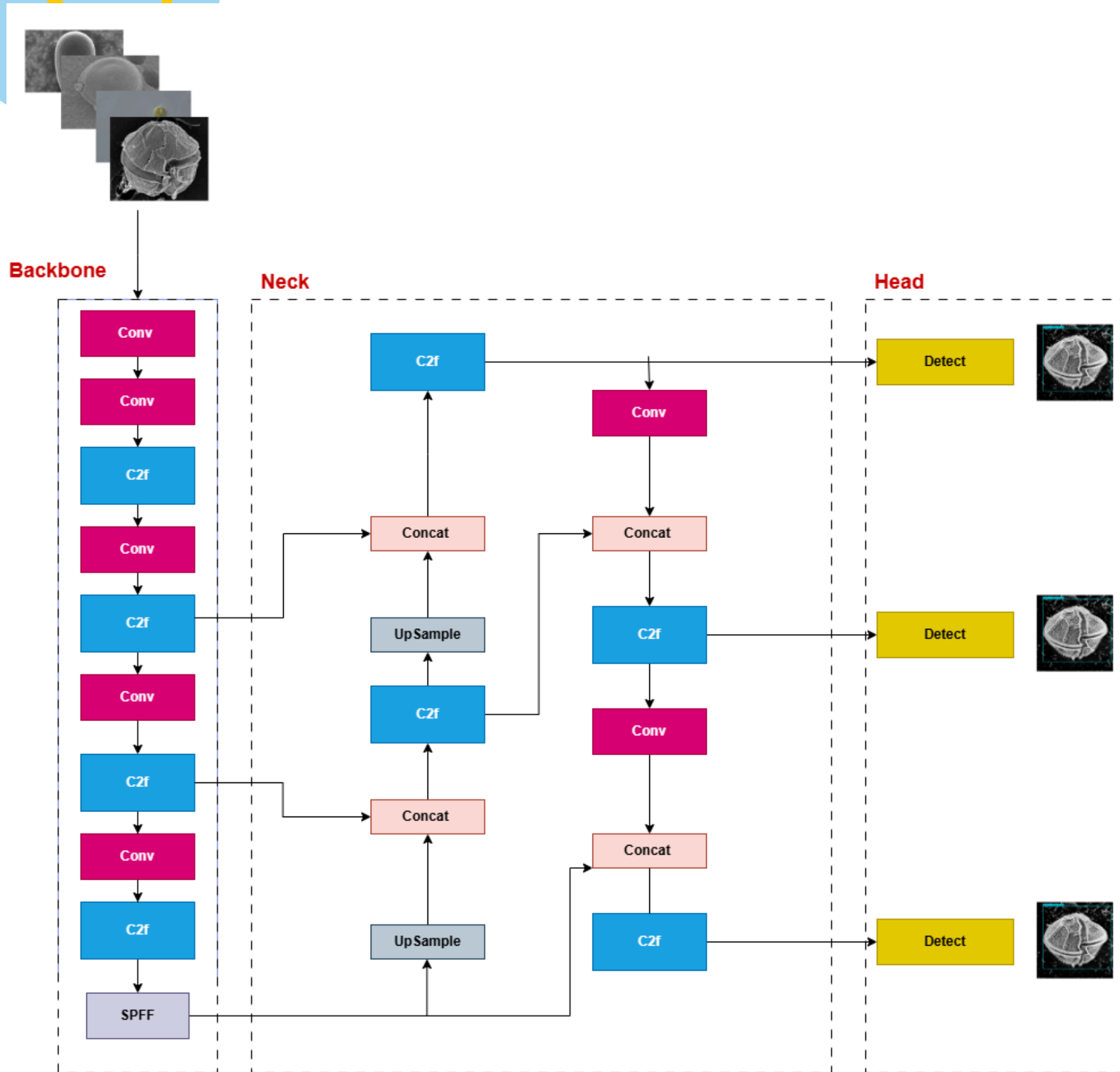
1. Enable skip connections for smooth gradient flow.
2. Support deep network training without vanishing gradients.
3. Extract increasingly abstract and hierarchical features.

✅ Key Advantages

- Residual Connections: Enable deep, stable training.
- Efficient Depth: Balances computational cost and representational power.
- High Precision: Effectively separates Alexandrium cells from other plankton types.
- Complements Detection Models: Strengthens overall analytical accuracy.



Yolo v8



◆ Backbone: Feature Extraction

Conv Layers : Capture spatial features from the input image.

C2f Modules : Efficient blocks with partial connections.

SPPF : Spatial Pyramid Pooling - Fast: Aggregates multi-scale spatial features for improved object detection.

◆ Neck: Feature Fusion

Upsample : Increases resolution to better combine features.

Concat : Merges feature maps from different layers.

C2f & Conv: Further refine features for richer representation.

→ Helps detect both small and large objects by combining shallow and deep features.

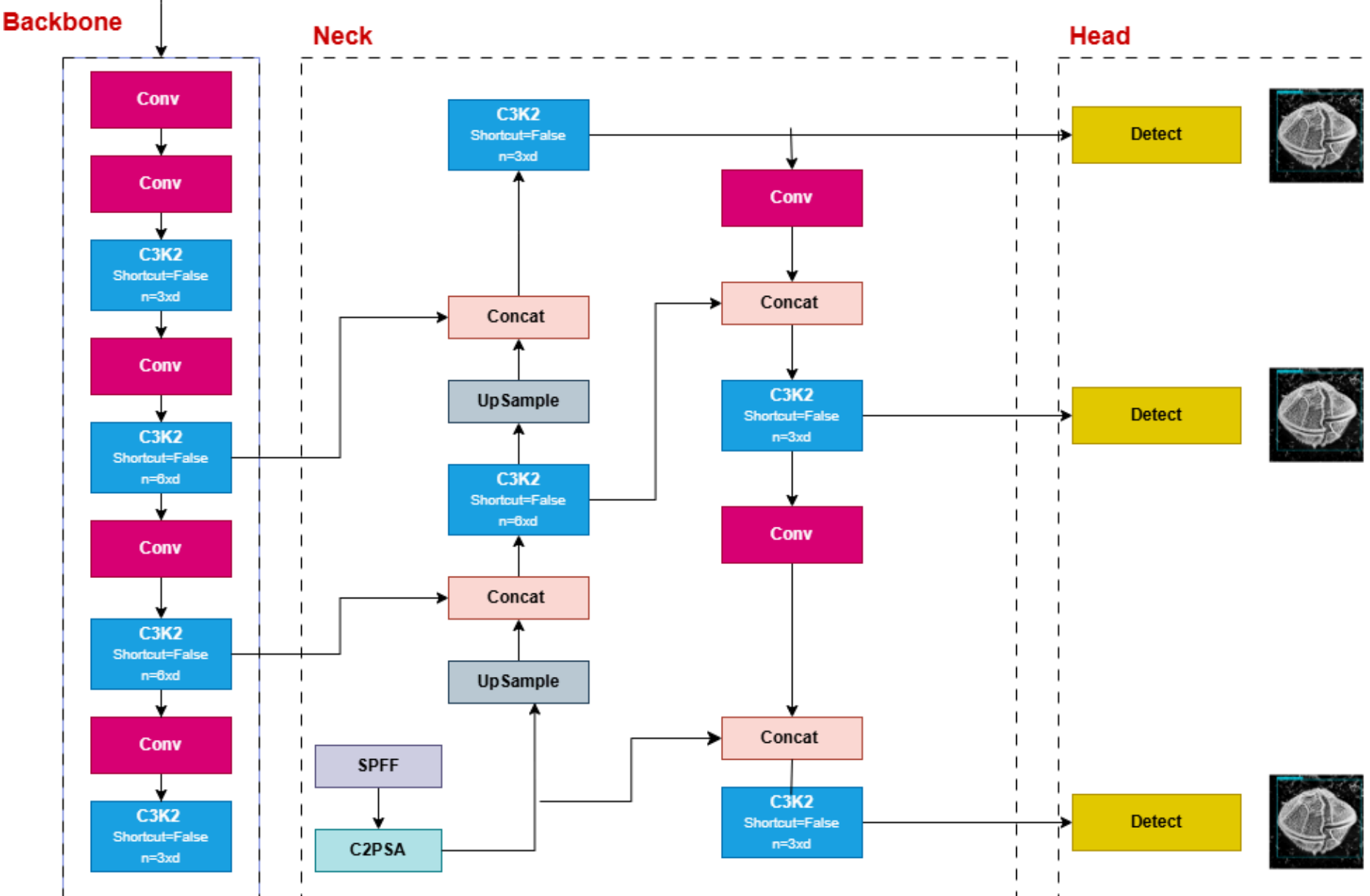
● Head: Detection Output

→ Output bounding boxes & class scores at 3 different scales.

✓ Key Strength:

- Supports accurate detection across varying object sizes.
- Optimized for speed and accuracy in complex scenarios.

yolo v11



◆ Backbone: Feature Extraction

Conv: Standard layers capturing low-level details.

C3K2 (Shortcut=False): Custom convolutional blocks (like YOLO's C3) with configurable depth (e.g., $3 \times d$, $6 \times d$): Progressively downsamples input while increasing feature abstraction.

◆ Neck: Feature Aggregation & Fusion

Upsample: Aligns feature map resolutions across layers.

Concat: Merges features across scales (FPN/PANet style).

C3K2: Refines fused features at each scale.

SPPF (Spatial Pyramid Pooling - Fast): Expands the receptive field for better scale-invariance.

C2PSA: Custom module for advanced attention-based feature refinement.

● Head: Multi-Scale Detection

→ Outputs bounding boxes & class scores at 3 different resolutions.

Multi-Scale Strategy

✓ Key Advantage:

- Tailored for complex scenes with varying object sizes.
- Combines efficiency, scale awareness, and robust detection performance.

RF-detr

Image Feature Maps

- The input image is converted into multi-resolution feature maps.

Encoder: Adaptive Visual Encoding

- Uses multi-scale deformable self-attention.
- Dynamically focuses on important regions.
- Flexible to scale and perspective variations.

Decoder: Object Query Refinement

- Operates on a fixed set of object queries.

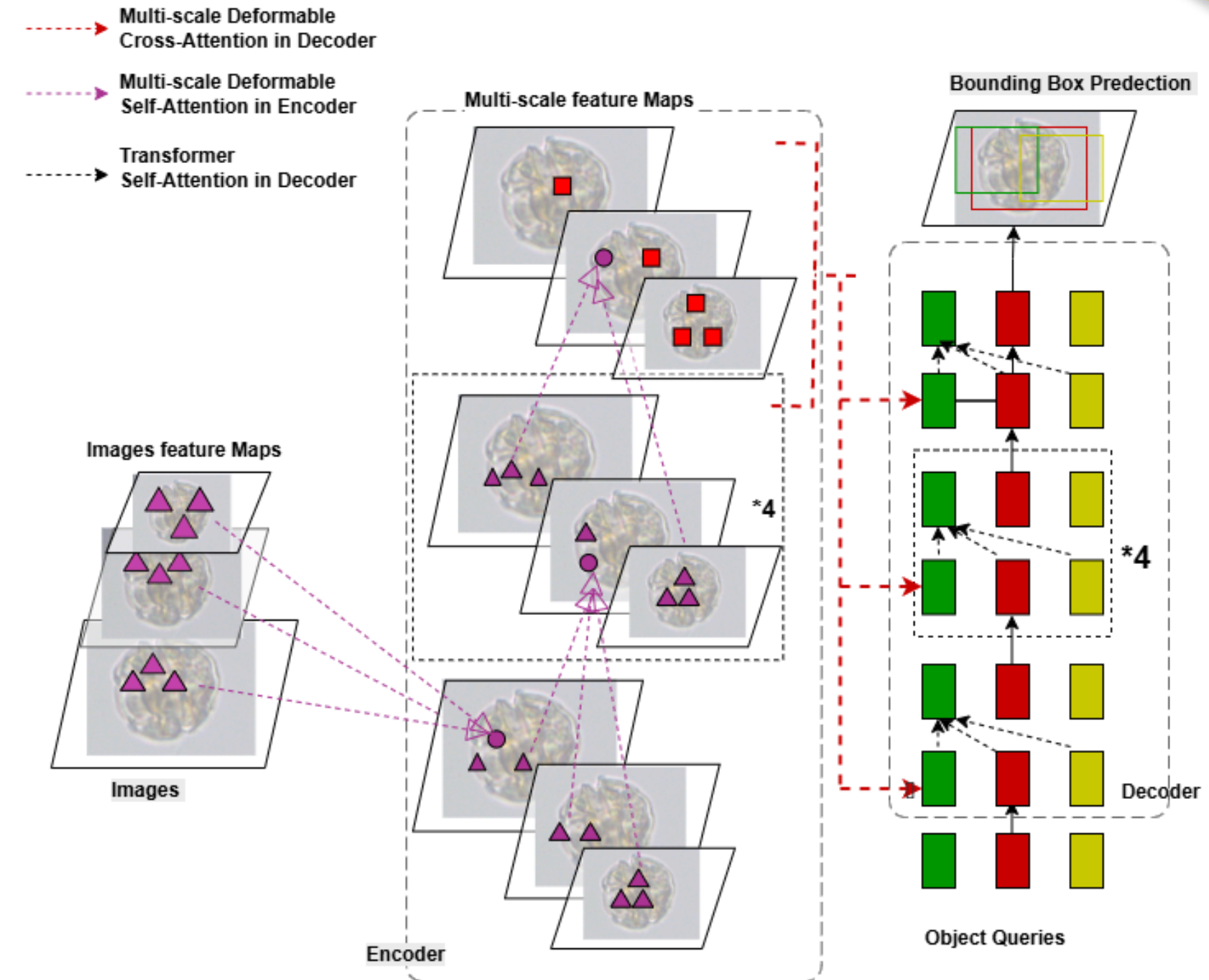
Two key mechanisms:

● Multi-Scale Deformable Cross-Attention: Aligns queries with relevant features across resolutions.

● Self-Attention: Enables interaction among queries to capture inter-object dependencies.

Bounding Box Prediction

- Refined queries are converted into:
 1. Bounding Boxes
 2. Class Labels
- Delivers accurate and robust detection results.



Key Strengths

- Global reasoning across the entire image.
- Long-range dependency modeling.
- Multi-scale attention enhances performance in cluttered or complex scenes.

Component	YOLOv8	YOLOv11	RF-DETR
Backbone	- Conv layers for spatial feature extraction - C2f modules with cross-stage partial connections for efficient learning - SPPF: Aggregates multi-scale spatial features	- Conv layers for low-level features - C3K2 blocks (Shortcut=False) for deeper residual learning (depth varies) - SPPF included for multi-scale representation	- Image feature maps generated via hierarchical attention - Encoder with multi-scale deformable self-attention to adaptively focus on important regions
Neck	- Upsample: Increases spatial resolution - Concat: Fuses features from different layers - C2f and Conv layers further process features	- Upsample + Concat (FPN/PANet-style fusion) - C3K2 blocks for refining fused features - SPPF + C2PSA module (Cross-Stage Partial Self-Attention) for enhanced context capture	- No traditional "neck" - Features passed directly from encoder to decoder using attention- multi-scale reasoning handled internally in attention mechanism
Head	- Detect blocks at 3 resolutions (small/medium/large) - Outputs bounding boxes & class probabilities	- Multi-branch Detect module at 3 scales - Outputs class probabilities & bounding boxes with improved accuracy	- Decoder refines fixed set of object queries using: *Cross-attention *Self-attention *Final bounding box prediction + class labels
Detection Approach	CNN-based with hierarchical feature fusion	CNN-based with enhanced residual depth and partial attention	Transformer-based using global reasoning and attention
Strengths	Lightweight, fast inference, strong real-time detection	Balanced depth & precision, more expressive features with attention modules	Handles complex, cluttered scenes with global attention and contextual reasoning
Multi-scale Support	Yes – SPPF + feature fusion support detection of varied object sizes	Yes – Enhanced multi-scale representation + self-attention	Yes – Handled via deformable attention across multiple scales

Component	YOLOv8	YOLOv11	RF-DETR
Backbone	Conv layers, C2f modules, SPPF for multi-scale spatial features	Conv + C3K2 blocks (Shortcut=False), SPPF, deeper residual learning	Multi-scale deformable self-attention encoder; spatial hierarchies
Neck	Up-sample, Concat, C2f and Conv layers	Up-sample, Concat, C3K2, SPPF, C2PSA module	No traditional neck; multi-scale reasoning via attention
Head	Detect blocks at 3 scales; bounding boxes and class probabilities	Three-scale Detect head; improved multi-scale precision	Decoder with deformable cross-attention and self-attention; predicts boxes and labels
Detection Approach	CNN-based hierarchical fusion	CNN with attention modules and deeper paths	Transformer-based with global attention
Multi-scale Support	Yes – SPPF + feature fusion	Yes – Multi-scale with attention	Yes – Via deformable attention mechanisms
Strengths	Lightweight, fast, real-time detection	Balanced accuracy and depth with partial attention	Excels in cluttered scenes; strong contextual reasoning

Component	ResNet18	ResNet50
Depth	18 layers	50 layers
Building Block	Basic-Block (2 layers)	Bottleneck Block (3 layers)
Conv1	7x7 conv, 64 filters, stride 2	7x7 conv, 64 filters, stride 2
Stage 1	2 x Basic-Block (64 filters)	3 x Bottleneck (256 filters)
Stage 2	2 x Basic-Block (128 filters)	4 x Bottleneck (512 filters)
Stage 3	2 x Basic-Block (256 filters)	6 x Bottleneck (1024 filters)
Stage 4	2 x Basic-Block (512 filters)	3 x Bottleneck (2048 filters)
Strengths	Lightweight, fast inference	Deeper features, better accuracy

Results for dataSet 1:

Detection results

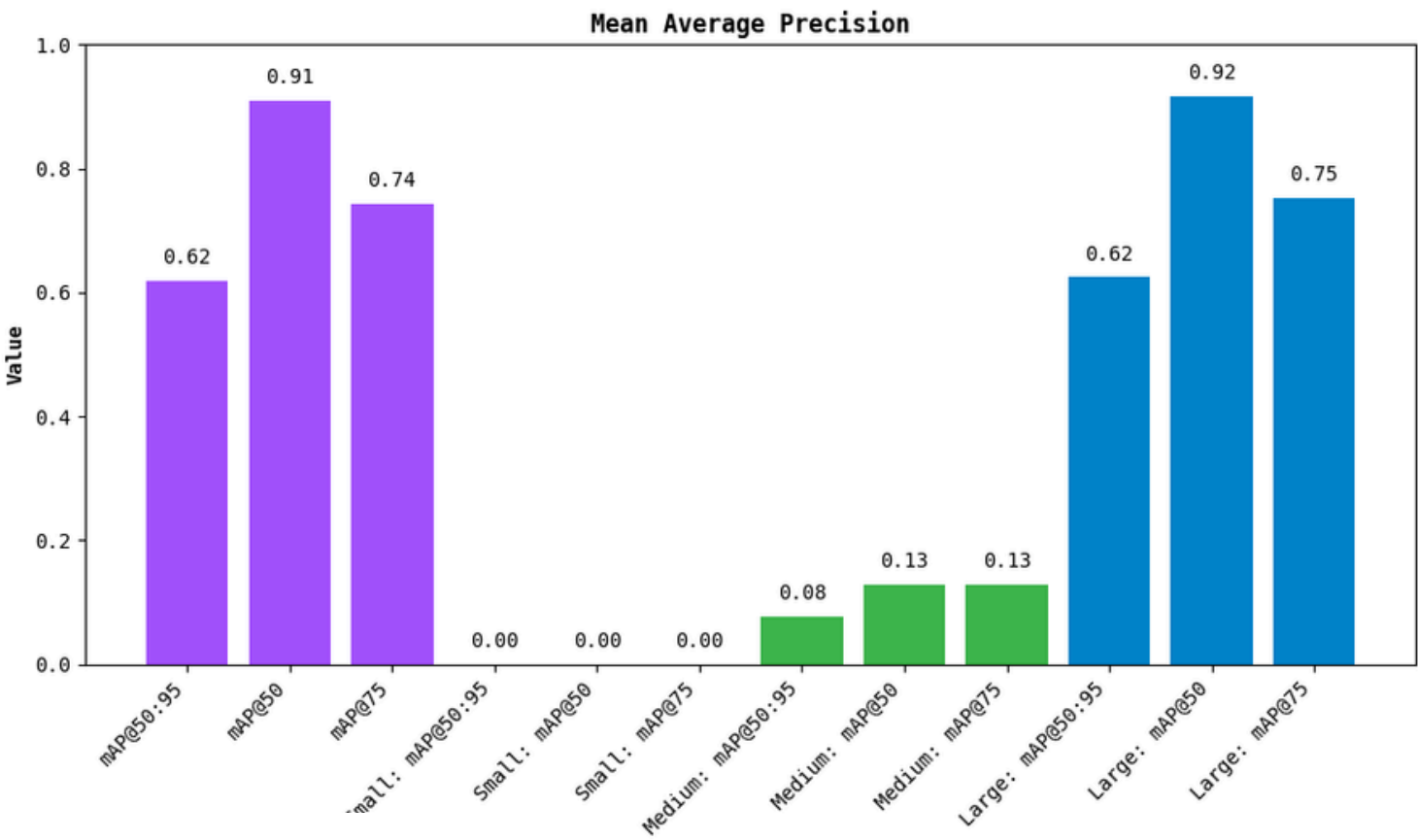
Model	mAP@0.50	mAP@0.50:0.95	Conclusion
YOLOv8	0.89	0.63	High object detection performance, but less accurate localization.
YOLOv11	0.87	0.62	Similar to YOLOv8, suitable for fast detection with moderate accuracy.
RF-DETR	0.80	0.73	Best for accurate localization and spatial precision, despite lower mAP@0.50.
<u>Conclusion:</u> Use YOLO for speed, RF-DETR for accuracy.			

Classification results

Model	Accuracy	Precision	Recall	Conclusion
YOLOv8	0.64	0.71	0.89	High recall; detects most instances but with more false positives.
YOLOv11	0.65	0.97	0.67	High precision; fewer false positives but may miss many true cases.
RF-DETR	0.83	0.83	0.87	Balanced performance; best overall for small or imbalanced datasets.
<u>Conclusion</u> RF-DETR is most reliable for classification.				

Phase	Description	Problem Identified	Results
Phase 1: Initial Setup	Used small versions of YOLOv8 and YOLOv11 with 70/20/10 data split.	Poor accuracy and inconsistent detection performance.	YOLOv8S: 83%, YOLOv11S: 80%
Phase 2: Data Augmentation & Label Verification	Applied augmentation techniques , manually corrected labels and change images sizes till we had seven version .	No significant improvement despite better data quality.	No major performance gain observed.
Phase 3: Pipeline & Training Strategy Modification	Adjusted input resolution, froze initial layers, resumed training from checkpoint, extended training with all layers and change the models into medium version .	Models struggled with low-res microscopic images.	YOLOv8M: 85%, YOLOv11M: 81%
Phase 4: Dataset Re-balancing	Identified class imbalance and redistributed samples for equal representation.	Large disparity between class distributions.	Balanced dataset created with 80/10/10 split.

final Results for dataSet 2:



Phase	Description	Problem Identified	Outcome
Phase 1: Initial Setup	Trained model with: 50 epochs, - batch size = 2, - learning rate = 1e-4.	Despite promising results, performance was limited by small batch size and under-optimized hyperparameters.	mAP@50 = 0.91, mAP@75 = 0.74, mAP@50:95 = 0.62
Phase 2: Hyperparameter Tuning Attempt	Increased batch size to 4	Technical limitation: Inadequate GPU memory and compute prevented training with higher batch size.	Training could not proceed as planned; model evaluation halted due to hardware constraints.

final Results for dataSet 2:

Phase 5 :Final Attempt and Best Results

Model Parmeters :

Parameter	YOLOv8 (Medium)	YOLOv11 (Medium)
Epochs	100	100
Image Size	960	960
Batch Size	16	16
Optimizer	AdamW (Auto)	AdamW (Auto)
Learning Rate (lr)	0.001667	0.001667
Momentum	0.9	0.9

final Results for dataSet 2:

Phase 5 :Final Attempt and Best Results

detection results

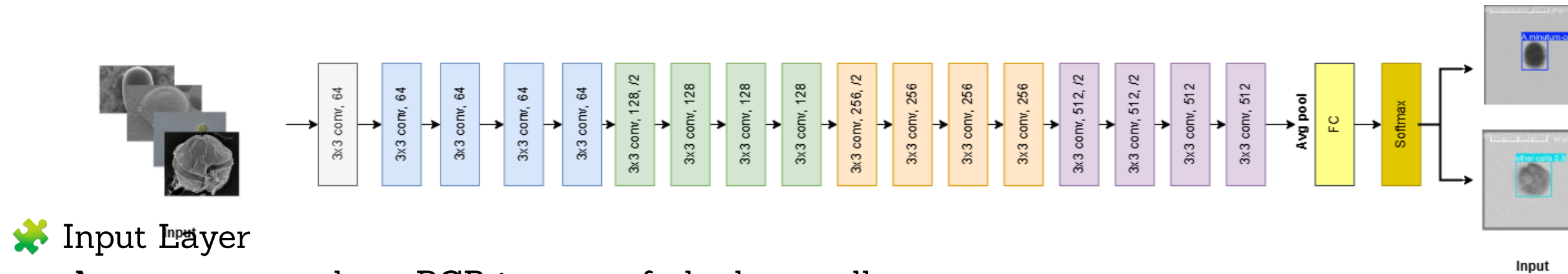
Model	mAP@50	mAP@50:90	Conclusion
YOLOv8	0.93	0.59	Strong detection performance, slightly less precise at higher Intersection over Union.
YOLOv11	0.93	0.60	Matches YOLOv8 at basic detection, but more consistent under stricter localization.
Conclusion: YOLOv11 is preferred when accurate object boundaries are critical.			

Classification results

Model	Accuracy	Precision	Recall	Conclusion
YOLOv8	90.5%	0.87	0.87	Strong balance , good overall performance.
YOLOv11	93.0%	0.89	0.86	Higher accuracy and precision, slightly lower recall.
ResNet18	99.4%	0.992	0.9937	Highest performance across all metrics.
ResNet50	98.1%	0.9849	0.9828	Excellent results, just below ResNet18.
Conclusion: ResNet18 is the most accurate and reliable classifier.				

Resnet 18

Output Classes:
◆ Alexandrium-cells ◆ Other-cells



✚ Input Layer

- Accepts grayscale or RGB images of plankton cells.
- Starting point for feature extraction.

● Initial Convolution

- 3×3 Conv, 64 filters :Captures low-level features (e.g., edges, textures).

● Stage 1

- Three 3×3 Conv layers, 64 filters:Deepens early feature representations.

● Stage 2

- 3×3 Conv, 128 filters, stride 2:Downsampling with feature depth expansion.
- Two 3×3 Conv layers, 128 filters:Mid-level abstraction.

● Stage 3

- 3×3 Conv, 256 filters, stride 2:Further spatial reduction.
- Two 3×3 Conv layers, 256 filters: Captures complex patterns.

● Stage 4

- 3×3 Conv, 512 filters, stride 2: Final spatial downsampling.
- Two 3×3 Conv layers, 512 filters: Extracts high-level, class-specific features.

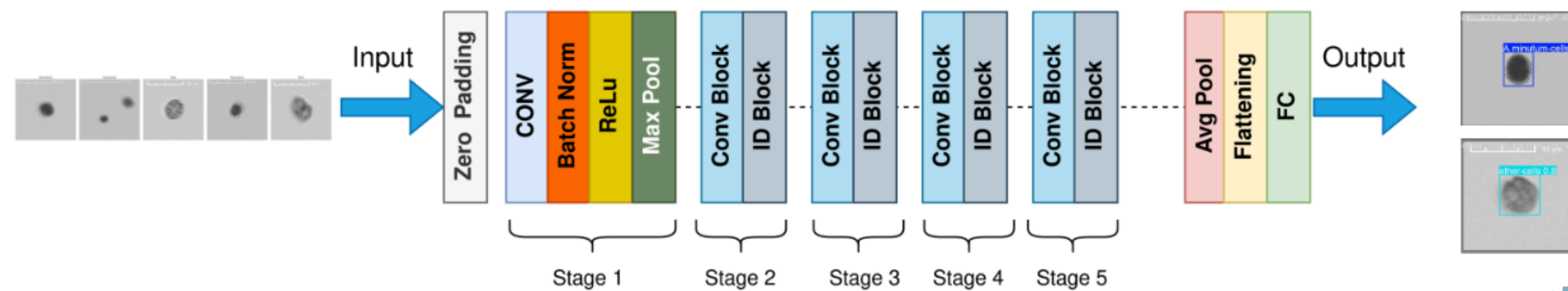
⚙️ Post-Convolution Layers

- Average Pooling: Reduces feature maps to single values.
- Fully Connected Layer: Outputs class scores.
- Softmax: Converts scores into class probabilities.

✓ Key Benefits

- Hierarchical feature extraction.
- Residual connections stabilize deep training.
- Effective generalization across complex image data.

Rsent50



Input & Preprocessing

- Input Layer:
- Receives multiple grayscale or RGB images of plankton cells.
- Preprocessing:
- Zero Padding: Maintains spatial dimensions during convolutions.

◆ Stage 1: Initial Feature Extraction

- Convolutional Layer: Captures basic visual features (edges, textures).
- Batch Normalization: Stabilizes and accelerates training.
- ReLU Activation: Enables learning of complex patterns.
- Max Pooling: Downsamples feature maps, preserving key features.

◆ Stages 2–5: Residual Learning:

1. Enable skip connections for smooth gradient flow.
2. Support deep network training without vanishing gradients.
3. Extract increasingly abstract and hierarchical features.

Post-Convolution Layers

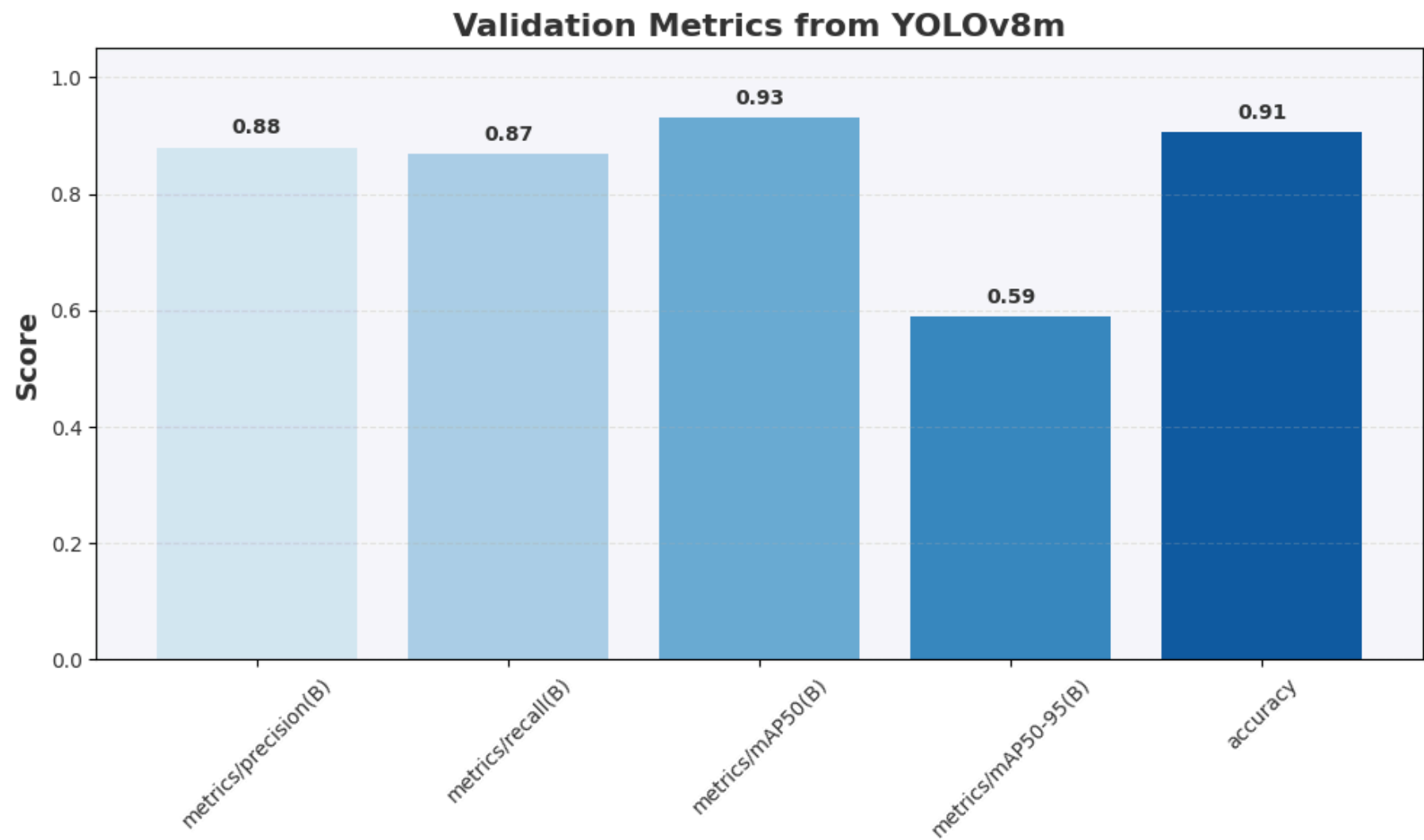
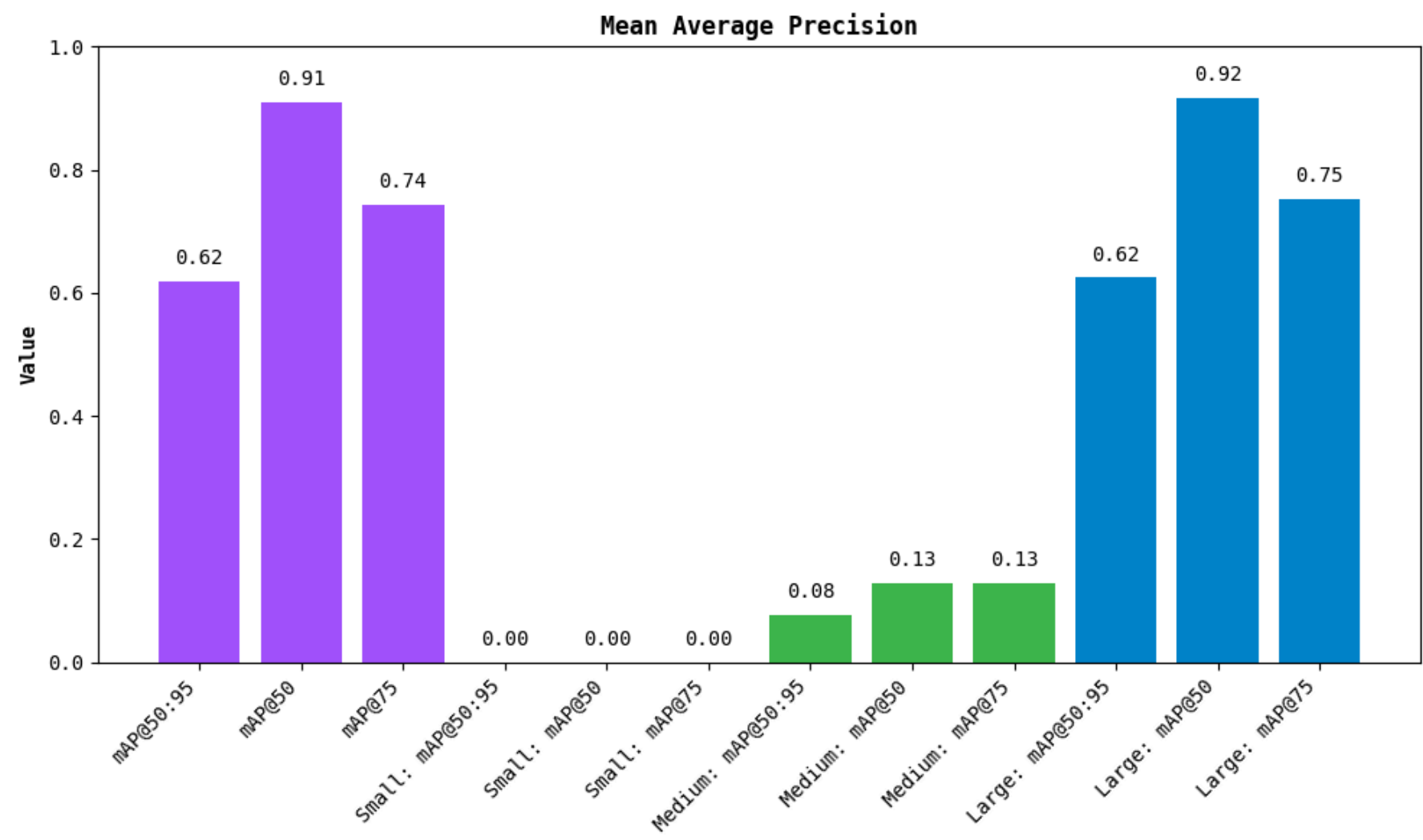
- Average Pooling: Summarizes each feature map into a single value.
- Flattening: Converts 2D maps into a 1D feature vector.
- Fully Connected (FC) Layer: Performs final classification.

Output

- ◆ “Alexandrium-cells”
- ◆ “Other-cells”

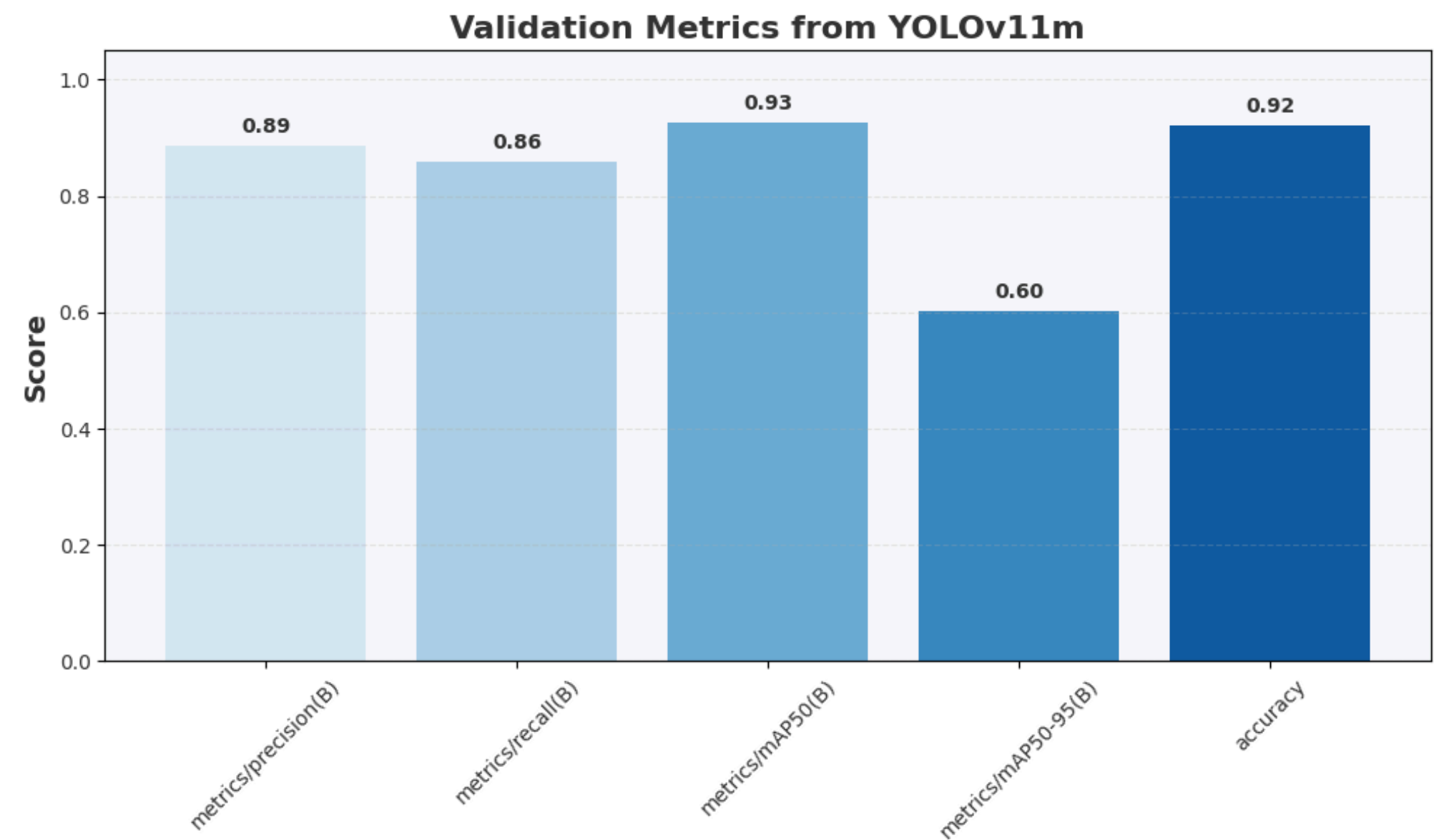
Key Advantages

- Residual Connections: Enable deep, stable training.
- Efficient Depth: Balances computational cost and representational power.
- High Precision: Effectively separates Alexandrium cells from other plankton types.
- Complements Detection Models: Strengthens overall analytical accuracy.



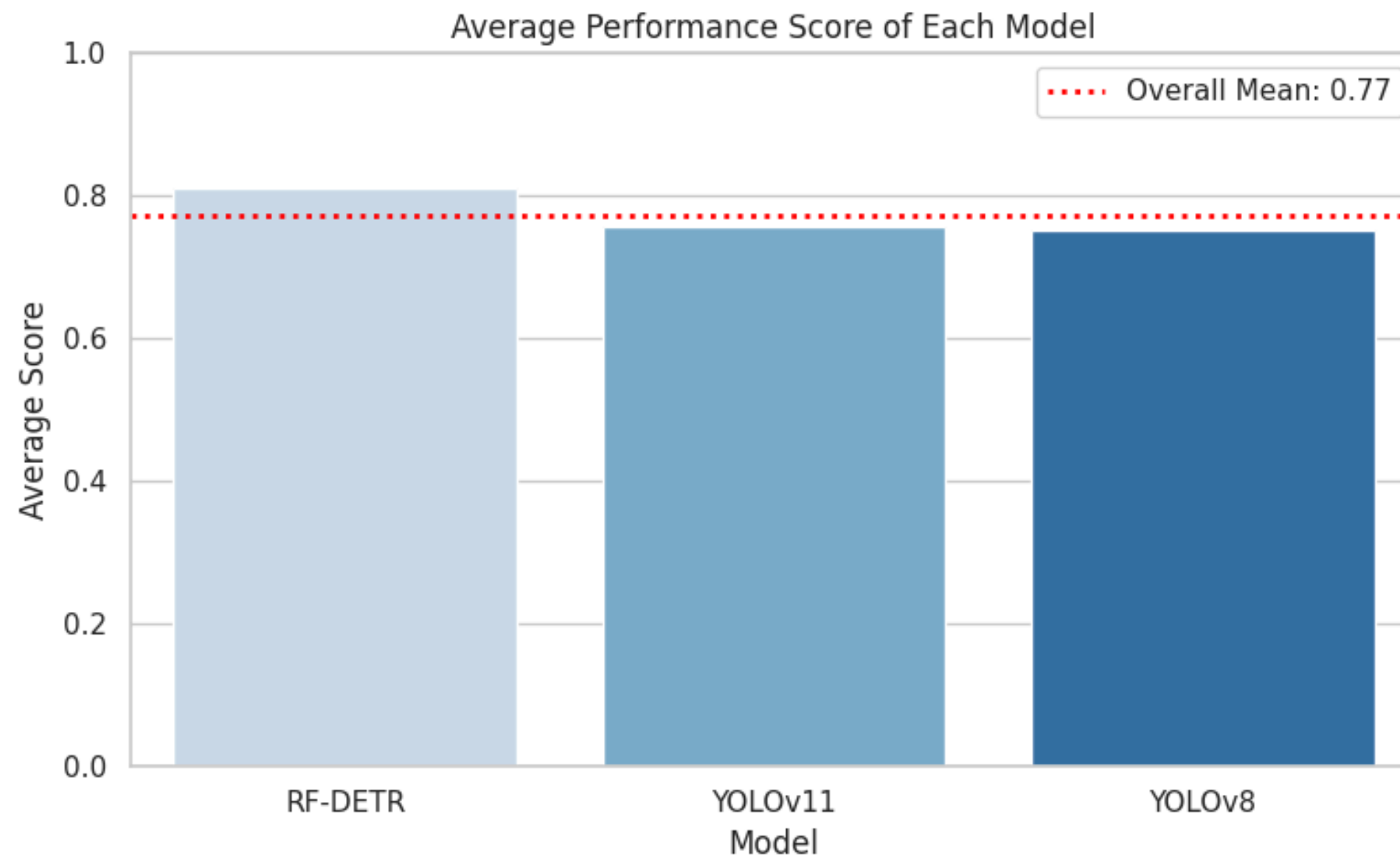
Test Classification Report:					
	precision	recall	f1-score	support	
	0	0.9934	0.9836	0.9885	305
	1	0.9752	0.9899	0.9825	199
accuracy			0.9861		504
macro avg	0.9843	0.9868	0.9855		504
weighted avg	0.9862	0.9861	0.9861		504

Resnet18

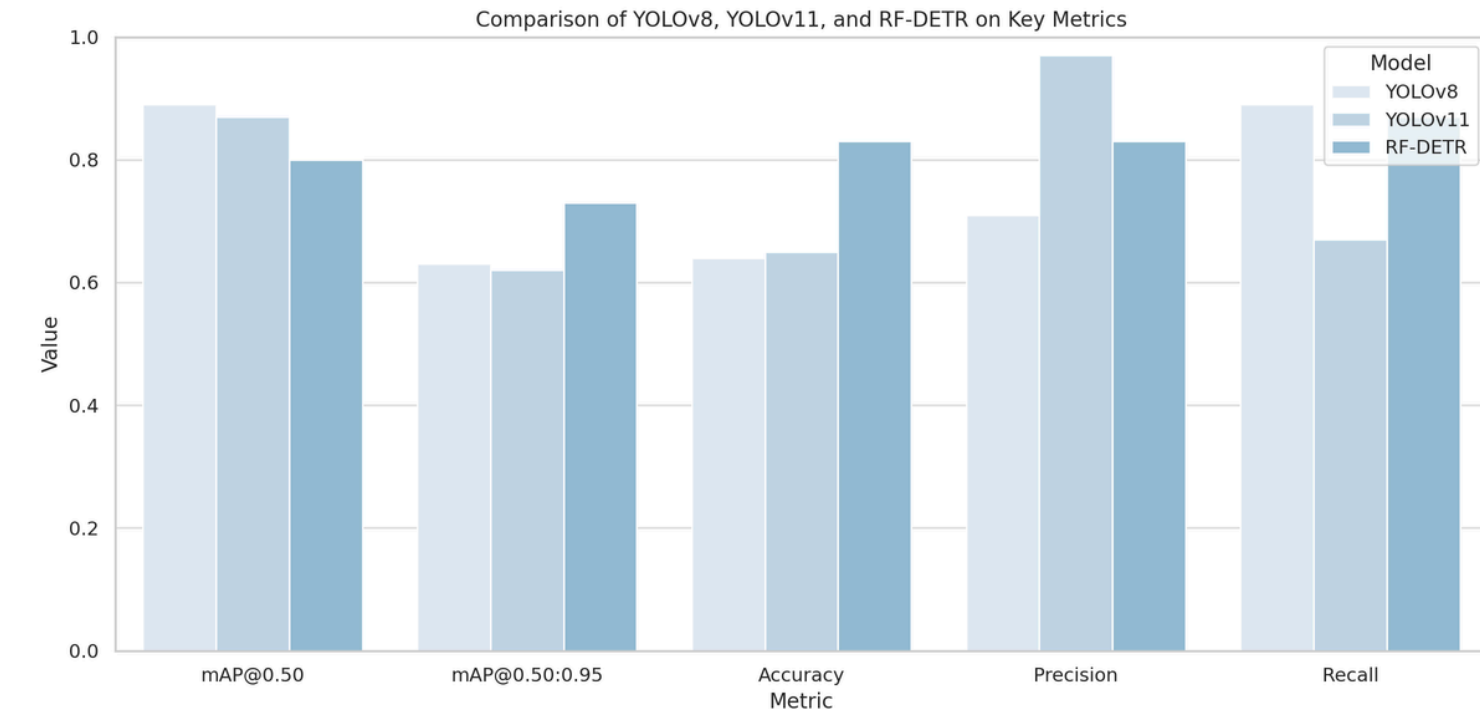
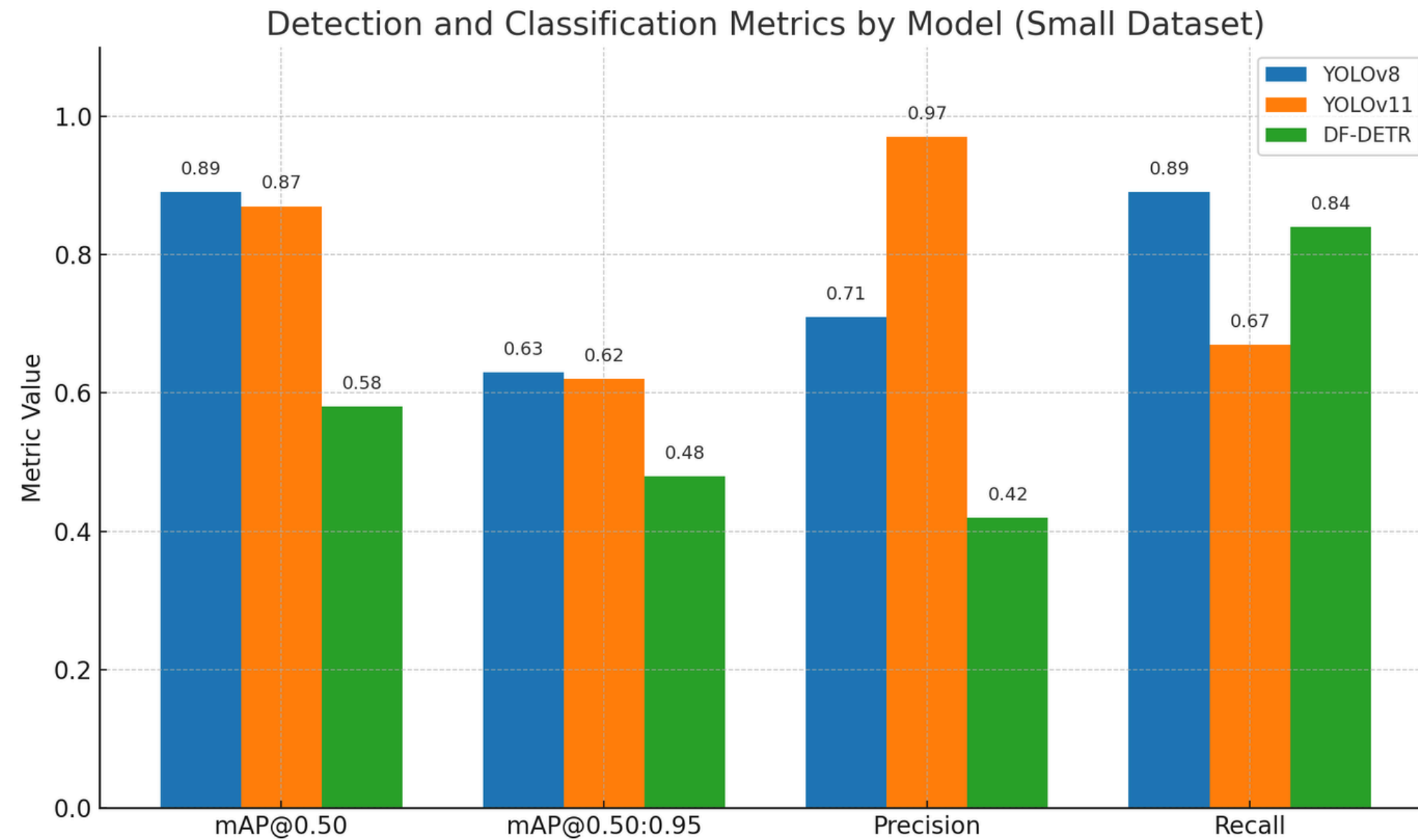


Test Classification Report:				
	precision	recall	f1-score	support
0	0.9806	0.9967	0.9886	305
1	0.9948	0.9698	0.9822	199
accuracy			0.9861	504
macro avg	0.9877	0.9833	0.9854	504
weighted avg	0.9863	0.9861	0.9861	504

REsnet 50



**Houni rit baad heki el conclution eli
fe table naamlou hedhi tconfirmha
akthar w zid hedhi nejmou
nekht mou biha chnou el conclusion
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:the rf-detr was the best so that
why we wanted to apply it on our
dataset 2**



**Hedhi hal nhtouha maa el
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CRITÈRES DE PERFORMANCE

Sensibilité
(rappel)

$$\frac{TP}{TP + FN}$$

- Le rapport entre le nombre total de patients positifs correctement classés et le nombre total de patients positifs

Exactitude

$$\frac{TP + TN}{TP + TN + FP + FN}$$

- Le taux d'images correctement classées

Précision

$$\frac{TP}{FP + TP}$$

- Le rapport entre le nombre total de catégories positives correctement classées et le nombre total de catégories positives prédites

F1-score

$$2 \frac{\text{precision} \times \text{rappel}}{\text{precision} + \text{rappel}}$$

- Une mesure qui combine la précision et le rappel d'un modèle de classification